Systematic Trading Strategies

Where durable returns come from, how an individual can run them, and how to tell an edge from a lucky backtest

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Abstract

A systematic strategy is a rule, fixed in advance, for what to hold. The rules that keep earning after they become widely known are those that pay investors to bear something unpleasant. This chapter covers where such returns come from, which strategies survive, and how to evaluate them.

A strategy's return splits into a static part, from holding its average position, and a timing part. The static part is usually most of the return, and it is cheap to buy. A rule that holds US stocks only while the market is above its 10-month average earned about six-sevenths of its 99-year return from being invested three-quarters of the time. Its timing skill was too small to prove in a century. It did cut the worst drawdown in half. Before costs, trading is zero-sum, so every lasting excess return needs a counterparty who keeps accepting the worse side. Returns paid by buyers of crash protection, by institutions with forced trades, or by repeated behavioral errors tend to last. Returns from noticing something first are competed away. Published stock-market anomalies lost about 58% of their returns after publication. Lasting premia must cause pain. Value stocks have not regained their 2007 high by the end of 2025, and momentum spent 24 years below its 1932 peak.

Six families have survived: a rebalanced mix of broad markets, multi-market trend following, value combined with momentum, carry, low volatility, and option selling. Calendar and news-reaction effects have mostly disappeared. An individual's advantage is the absence of benchmark and career pressure, but most investors trail their own funds by one to several points a year through badly timed flows. Taxes and costs can outweigh the strategy itself. An 8% pre-tax return compounds at between 4.7% and 8% after tax, depending on the account and turnover. A diversified futures program needs about 140,000 dollars, even with micro contracts.

Backtests overstate results. The best of 100 skill-free rules tested on 10 years of data beats the stock market's century-long record. Confirming a good strategy with four-in-five probability takes about 30 years of history. A study of 888 retail algorithms found that backtests predicted almost nothing about live results. A good strategy loses money about one year in three. The chapter recommends naming who pays before testing anything, holding a few durable strategies that fail at different times, planning on half the backtested performance, and writing stop rules before starting.

What this is. A systematic trading strategy is a rule, fixed in advance, that turns data into positions. This chapter explains where such rules get their returns, and which of those returns survive being widely known. It explains how an individual can actually run a strategy from a retail brokerage account, and how to evaluate one honestly. It is written for a technically strong reader who intends to build something. So it prefers mechanisms to recipes, and failure modes to feature lists.

How to read this chapter. Four questions motivated it, and they map onto sections:

Question	Where it is answered
What are systematic strategies?	§1, which defines them and gives the identity that splits any strategy's return into parts; §6, which organises them
Which ones don't get traded away?	§2, on the economics of why a return persists; §5, which judges nine strategy families on durability
How do individual investors run them?	§7
How should they be evaluated?	§8

§3 (history) and §4 (the annotated bibliography) are reference, and §9 synthesises. Different readers can start in different places.

- Readers with time for only three things should read §1.5–§1.6, all of §2, and §8.3.
- Readers about to build should read §7 and §8 in full, and use §5 as reference.

Appendix A explains, from first principles and in dependency order, every concept that the main text relies on without stopping to explain.

Objectives. After this chapter, you should be able to:

- split any strategy's expected return into a static part, a timing part and cost, and say which part a given strategy earns;
- name who pays for a return, classify it as a risk, structural, behavioural or informational premium, and judge whether it will survive being widely known;
- describe the main strategy families, how each one fails, and how an individual can run it;
- estimate the effect of costs, taxes, account type and capital on a strategy's net return;
- judge how much a backtest can show, correct it for the number of variants tried, and set a realistic expectation for live performance;
- size and combine strategies, and write stopping rules before going live.

Relationship to the other chapters. This chapter is the umbrella over several deeper ones. Momentum in Financial Markets and Trend-Following in Financial Markets treat two of the strategy families of §5 in depth. Portfolio Construction and the Covariance Matrix gives the detail behind sizing and combining strategies (§7.8). Foundations of Econometrics gives the detail behind the statistics of §8. Implied Volatility and Dealer Hedging and Gamma Exposure are the background to volatility selling (§5.7). Market Regimes and Hidden Markov Models is the background to regime filters and purged cross-validation. Each chapter stands alone. This one shows which of them a reader needs.

Where the numbers come from. Numbers described as a calculation run for this chapter come either from the Kenneth French data library or from simulations. The generators are in `figures/`. They illustrate mechanisms, and are not recommendations of anything tradeable.

A warning about scope. [Practice] Nothing here is investment advice. The retail implementation details, such as taxes, margin rules and account types, are for the United States as of September 2026. They change, and other jurisdictions differ materially. Where a source has a commercial interest in its own conclusion, the text says so.

Epistemic tags. The chapters in this collection flag claims by status:

- **[Fact]** — replicated across independent datasets or implementations; broad agreement.
- **[Contested]** — documented, but with live disagreement about magnitude, robustness, or cause.
- **[Hypothesis]** — a proposed mechanism, not decisively tested.
- **[Practice]** — practitioner convention; the evidence may be private or absent. A [Practice] claim is not a debunked one.

Tags appear only where the status changes what a reader should do. A tag governs the sentence or clause it opens. Untagged sentences are definitions, derivations or arithmetic: true by construction rather than by evidence.

Notation. The table lists every symbol that recurs in the chapter, with the section that defines or first uses it. Symbols used in only one section are defined where they appear.

Symbol	Meaning	Defined in
t	Decision date; a return indexed $t+1$ is earned over the following period $(t, t+1]$	§1.2
N; i	Number of assets; asset index	§1.2
$r_{i,t+1}$	Asset i's **excess return**: its return minus the cash (T-bill) rate, so a strategy that holds nothing earns zero excess return	§1.2
$w_{i,t}$	Position in asset i chosen at t, as a fraction of capital: positive for long, negative for short; $\sum_i \lvert w_{i,t}\rvert > 1$ means leverage	§1.2
$\mathcal{I}_t$; θ	Information available at t; parameters fixed in advance, so that a strategy is a rule $w_t = f(\mathcal{I}_t; \theta)$	§1.2
π_{t+1}; c_{t+1}	Strategy's excess return after costs; trading cost as a fraction of capital	§1.2
ν; δ	**Static** (passive) and **timing** (active) components of expected return	§1.5
μ; σ; ρ	Expected excess return; volatility; correlation; all annualised unless stated	A.1, A.2

Symbol	Meaning	Defined in
SR; $\widehat{\mathrm{SR}}$	Annualised **Sharpe ratio** μ/σ; its estimate from data	§1.1, A.3
IC; BR; TC; IR	Information coefficient; breadth; transfer coefficient; information ratio	§1.7
g	Compound growth rate	§5.1, §7.8, A.1
$\sigma^\star$	Volatility target	§5.2, §7.4
κ; u	One-way trading cost per unit traded; annual one-way turnover as a multiple of capital	§8.4, A.7
τ	Tax rate	A.29
T; n; q	Backtest length in years; number of observations; observations per year ($q = 12$ for monthly data, about 252 for daily)	§8.2
M	Number of strategy variants tried before one is chosen	§8.3
γ_3; γ_4	Skewness; kurtosis ($\gamma_4 = 3$ for a normal distribution)	§8.2, A.4
Φ; Φ^{-1}	Standard normal distribution function; its inverse	§8.2

Several symbols carry a qualification:

- For a single asset, the subscript i is dropped.
- Percentages written as $\%$ inside formulas are per year unless stated.
- In §6.3 and A.24, τ_e is an option's time to expiry, distinct from the tax rate τ.

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- A. Concepts and prerequisites — every idea the main text leans on without fully explaining, built from first principles and ordered by dependency.

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1. What a systematic strategy is

1.1 The wrong intuition

Most people first meet systematic trading as a search problem. On this view, the history of prices contains patterns that predict what comes next. A good enough search finds them. A strategy is a pattern plus the discipline to trade it. The work is mining, the tool is the backtest, and a better search makes a better strategy.

Each part of that picture misleads, and much of this chapter explains why.

- **Search finds noise before it finds signal.** Try 100 variants of a rule with no skill at all on 10 years of data. The best of them will show a Sharpe ratio of about 0.8. That is nearly double the 0.45 that the US stock market earned over the last century. §8.3 derives this, and the number comes from a calculation run for this chapter.
- **Patterns that are merely predictable get competed away.** Suppose a rule earns money without risk, cost or discomfort. Other people find it and trade it until it stops working. [Fact] The typical published anomaly in US stock returns lost about 58% of its return after publication (McLean & Pontiff, 2016; §2.3).
- **Most of what a durable strategy earns is not prediction at all.** It is payment for holding something that other people would rather not hold: risk, illiquidity, crash exposure or an unpopular position. The rule is a disciplined way of holding it. §1.6 examines one of the best-known retail timing rules. About six-sevenths of its 99-year return came from being invested, not from timing.

The right picture is this: **a systematic strategy is a fixed rule for deciding what to hold, and its long-run return is whatever the market pays for holding that, minus what it costs to trade.** The rule is a delivery mechanism. The return comes from somewhere else. The question that matters is where, and §2 is built around it.

1.2 Definition, and the anatomy of a strategy

Formally, a systematic strategy is a function, fixed in advance, from the information available at time t to positions:

$$w_t = f(\mathcal{I}_t;\, \theta).$$

Here w_t is the vector of positions held from t to $t+1$. $\mathcal{I}_t$ is everything knowable at t: prices, volumes, company accounts, futures curves, option prices and the calendar. θ collects the parameters chosen before the rule is run, such as look-back lengths, thresholds and a volatility target. The strategy's excess return over the next period is

$$\pi_{t+1} = \sum_{i=1}^{N} w_{i,t}\, r_{i,t+1} \;-\; c_{t+1},$$

where c_{t+1} is the cost of the trades that moved the book from w_{t-1} to w_t, charged to the period they fund.

One property makes a strategy *systematic*: f is fixed before the data it acts on arrive. So the same inputs always produce the same positions. That property, and only that one, makes a strategy testable. It allows the question of what the rule *would have done* in the past, without hindsight leaking into the answer. The same property also makes a strategy dangerous. A rule can be re-fixed after every look at the past, and a rule re-fixed 100 times is no longer being tested (§8.3).

In practice f breaks into five stages. Naming them helps, because each stage has its own characteristic way of failing:

Stage	Question it answers	Typical choices	Where it goes wrong
Universe	What may be held?	S&P 500 stocks; 20 liquid futures; 8 ETFs	Testing on today's survivors
Signal	What is believed about each asset?	Past return; valuation ratio; futures basis; implied minus realised volatility	Overfitting; look-ahead in the data
Sizing	How much of each?	Equal weight; inverse volatility; an optimiser; a volatility target	Estimation error; hidden leverage
Execution	How are the positions reached?	Market-on-close orders; limit orders; rebalancing bands	Costs; fills that never happen
Overlay	What stops a blow-up?	Exposure caps; drawdown rules; a kill switch	Rules never tested, or that cut risk at the bottom

Most of the published argument in this field concerns the second row. Most of the money lost by people running strategies is lost in the other four.

1.3 What systematic is not

The word is used loosely, and several neighbouring terms get folded into it. Keeping them apart prevents a surprising number of confused decisions:

Term	What it is	What it is not
Systematic	A rule fixed in advance, reproducible from the same data	Automated. A quarterly rebalance done by hand from a spreadsheet is fully systematic
Systematic	A property of the decision process	A level of sophistication. A 10-month moving average is as systematic as a neural network
Passive investing	Holding market weights, the one strategy that needs no forecast	The opposite of systematic. It is a systematic strategy whose rule is "hold the index"
Algorithmic execution	A rule for *how* to fill an order (VWAP, TWAP)	A rule for *what* to hold
High-frequency trading	A systematic subset at sub-second horizons, earning spreads and speed	Something an individual can compete in. It is out of scope here
Technical analysis	A family of price-based signals	Necessarily systematic. Chart reading is discretionary
Machine learning	One way to estimate f	A source of return. It finds whatever the data contain, including noise
A backtest	A measurement of what a rule would have done, under stated assumptions	Evidence that the rule will work (§8)

1.4 Why be systematic at all?

Four benefits are real:

1. **Testability.** The past behaviour of a rule can be measured. The past behaviour of one's own judgment cannot, because memory is selective. Testability is also the main hazard, because it invites over-testing (§8.3).
2. **Consistency at the moment of decision.** A rule does not panic, get bored or anchor on its purchase price. [Fact] The documented losses of individual investors come mostly from decisions made in the moment. They trade too much (Barber & Odean, 2000). They pay for immediacy with aggressive orders (Barber, Lee, Liu & Odean, 2009). They buy after rises and sell after falls (Dichev, 2007). A rule removes those errors at the point of execution. It does not remove the larger error, which is abandoning the rule after a drawdown (§7.9).
3. **Breadth.** A rule that takes 10 minutes to apply to one asset takes about as long to apply to 1,000. §1.7 shows that breadth is where most of the value of any forecasting skill lives.
4. **Auditability.** When a systematic strategy loses money, it is possible to ask whether it did what it was designed to do. That question separates bad luck from bad design. Discretion rarely allows it to be answered.

Four costs are real too:

- **Rigidity.** A rule cannot know that something structural changed this time.
- **Model risk.** A rule encodes assumptions that its author may not know they made.
- **Crowding.** Rules published in papers are run by many people at once. In the August 2007 "quant quake", many equity market-neutral funds lost heavily in a few days as crowded positions were unwound together (Khandani & Lo, 2011).
- **False confidence.** A backtest looks like knowledge.

How do the two approaches compare in practice? [Contested] Harvey, Rattray, Sinclair & Van Hemert (2017) compare hedge funds that describe themselves as systematic or discretionary. They find broadly similar performance once exposure to well-known factors is adjusted for. The authors work for a systematic manager. Frazzini, Kabiller & Pedersen (2018) show that a leveraged, rules-based portfolio of cheap, safe, high-quality stocks reproduces much of Warren Buffett's record. These authors also work at a systematic firm. The defensible conclusion is not that rules beat judgment. It is that rules can express much of what good judgment does, and unlike judgment, rules can be tested.

1.5 The master decomposition

Take the expectation of π_{t+1}, and apply the identity $\mathsf{E}[XY] = \mathsf{E}[X]\,\mathsf{E}[Y] + \mathsf{Cov}(X, Y)$ to each term:

$$\mathsf{E}[\pi_{t+1}] = \underbrace{\sum_{i=1}^{N} \mathsf{E}[w_{i,t}]\ \mathsf{E}[r_{i,t+1}]}_{\nu\ \text{(static)}} + \underbrace{\sum_{i=1}^{N} \mathsf{Cov}(w_{i,t},\, r_{i,t+1})}_{\delta\ \text{(timing)}} - \mathsf{E}[c_{t+1}].$$

This is an identity, not a model. It holds for every strategy, with no assumptions about how prices behave. Lo (2008) introduced it to ask where active managers' returns come from. It is the first of the three ideas on which this chapter is built.

- ν, the **static** or passive component, is what holding the strategy's *average* portfolio permanently would earn. It requires no forecasting at all, only a willingness to hold the average position. It can usually be bought for the cost of a buy-and-hold portfolio.
- δ, the **timing** or active component, measures whether the rule holds more of an asset just before that asset does well. It is the only part of the return that requires predicting anything.
- The **cost** term is driven by turnover. It is the only term whose sign is known in advance.

Lo calls $\delta/(\nu+\delta)$ the *active ratio*. Three consequences make the decomposition worth computing for every strategy under consideration:

1. **It reveals what the strategy is paid for.** A strategy whose return is mostly ν is a premium-harvesting strategy in an active costume. Most of it can usually be bought more cheaply and with less tax (§7.2). A strategy whose return is mostly δ is a forecasting strategy. It needs the statistical scrutiny of §8, because δ is the component that noise can fake.
2. **Timing can be negative inside a profitable strategy.** Many active strategies earn a static premium and lose a little through their timing. Israelov & Nielsen (2015) show that the covered call is exactly this. It earns an equity premium and a volatility premium, both static. It loses a negative equity timing component, because the position's equity exposure falls as the market rises (§5.7).
3. **Timing is hard to measure.** δ is a covariance between a slowly moving position and a noisy return. Its t-statistic grows only with the square root of the sample length. §1.6 gives an example in which 99 years are not enough.

Two refinements matter in practice. First, the split is per asset. So in a cross-sectional strategy, "timing" includes rotation between assets. A stock that enters a momentum portfolio after rising contributes to δ. A permanent tilt toward small companies contributes to ν. The useful reading is *persistent tilt* against *time-varying bet*. That is also the split that decides turnover, and therefore cost. Second, the identity decomposes the **mean**. A rule that holds less when volatility is high can raise the Sharpe ratio and shrink drawdowns without timing the mean at all. It times the *second* moment. That effect is behind volatility-managed portfolios (Moreira & Muir, 2017) and much of why trend rules cut drawdowns (§5.2). It appears in the risk numbers, not in δ.

1.6 A worked instance: the 10-month moving average

Faber (2007) popularised a rule that many individuals still run. Hold the stock market when its price is above its 10-month moving average, and hold cash otherwise. Here the rule is applied to the US market and split with the identity of §1.5. The data are the Kenneth French library's monthly total returns. The signal is computed at each month-end on the total-return index and held for the following month. Cash earns the T-bill rate, and there are no costs. This is a calculation run for this chapter, and the generator is `figures/st_factor_history.py`.

What the rule earned, split into its static and timing parts:

Period	Time invested	Rule's excess return	Static ν	Timing δ	t of timing
1927–2025	73%	6.90%	5.94%	0.96%	1.2
1927–1962	68%	8.04%	6.71%	1.33%	0.7
1963–1999	76%	5.59%	5.44%	0.15%	0.1
2000–2025	74%	7.21%	5.44%	1.76%	1.3
1927–2025 without Jul 1929–Jun 1932	75%	7.25%	7.48%	−0.23%	−0.3

And what it did to risk, for the market and then for the rule:

Period	Sharpe ratio, market → rule	Worst drawdown, market → rule
1927–2025	0.44 → 0.55	−84% → −43%
1927–1962	0.43 → 0.56	−84% → −43%
1963–1999	0.47 → 0.46	−47% → −24%
2000–2025	0.47 → 0.70	−50% → −18%

Returns are annualised excess returns over T-bills. Drawdowns are measured on total-return wealth. The t-statistic is that of the slope in a regression of next month's market excess return on the rule's position. The sample starts in May 1927, once the moving average has 10 months of history. That is why the market's Sharpe ratio here is 0.44, not the 0.45 of §2.6.

Five lessons can be read off these tables, and each generalises:

1. **Most of the return is simply being invested.** The rule held the market 73% of the time. Holding 73% of the market permanently would have earned 5.94 of its 6.90 points a year. About six-sevenths of the headline return needed no timing skill at all.
2. **The timing component is not established in 99 years.** It is 0.96 points a year, with a t-statistic of 1.2. Over the same period, the market premium itself has a t-statistic of 4.5.
3. **What timing there is lives in a handful of episodes.** Remove the three years from July 1929, and the timing component turns slightly negative. It was essentially zero from 1963 to 1999. After 2000 it came from two bear markets, 2000–02 and 2008.
4. **What the rule did robustly was change the shape of the distribution.** The worst drawdown roughly halved in every period. The Sharpe ratio rose in 1927–1962 and 2000–2025, and was flat in 1963–1999. In that period, whipsaw losses in the rest of those decades used up what the rule saved in the 1973–74 bear market. Most of the improvement came from being out of the market while volatility was high. That is timing of the second moment, not prediction of the first.
5. **Trading costs are trivial here, but taxes are not.** About 1.5 switches a year cost almost nothing at retail spreads. But every exit realises accumulated gains, often short-term ones in a US taxable account (§7.2).

Does the rule "work"? That depends on what it was bought for. As a device for forecasting returns, the evidence is weak. As a rule that reliably cuts exposure during prolonged bear markets, it has done what it claims for a century. The price is small, repeated whipsaw losses in bull markets. That is the profile of insurance. It is why trend-following appears in §5.2 as a durable strategy, not as an anomaly. An investor who knows which of the two they bought can hold the rule through 1963–1999. One who does not tends to abandon it in year three.

1.7 Skill, breadth, and why diversified rules win

How much is a forecasting skill worth? The *fundamental law of active management* of Grinold (1989) answers with an approximation:

$$\text{IR} \;\approx\; \text{TC} \times \text{IC} \times \sqrt{\text{BR}}.$$

The **information ratio** IR is the Sharpe ratio of the active bets. The **information coefficient** IC is the correlation between the forecasts and the returns that follow. **Breadth** BR is the number of *independent* bets per year. The **transfer coefficient** TC was added by Clarke, de Silva & Thorley (2002). It is the correlation between the positions that the forecasts call for and the positions actually held. Constraints such as long-only rules or position limits push it below one.

Some numbers recalibrate intuition:

- **Skill is small.** An IC of 0.05 means that the forecast gets the direction right 51.6% of the time. (For jointly normal variables with correlation ρ, the probability that they share a sign is $\frac{1}{2} + \arcsin(\rho)/\pi$.) [Practice] Useful signals commonly have ICs between about 0.02 and 0.10.
- **One asset wastes it.** Consider timing a single index once a month, with no constraints on expressing the forecast (TC $= 1$). That gives BR $= 12$ and IR $= 0.05\sqrt{12} = 0.17$. This is a Sharpe ratio too small to confirm in a lifetime (§8.2), from a signal most people would consider good.
- **Breadth multiplies it.** The same IC applied monthly across 50 independent futures markets gives BR $= 600$ and IR $= 1.2$. That arithmetic, not a better signal, is why professional trend-followers trade 50 to 150 markets.
- **Breadth is rarely what it seems.** Across 500 stocks monthly, the formula gives BR $= 6000$ and IR $= 3.9$, which no one achieves. Stocks share common factors, so the effective number of independent bets is far smaller. Constraints drag TC down. Turnover costs grow with the number of bets.

For an individual, the cheapest available improvement is almost never a cleverer signal. It is more independent bets: more assets, and more strategies that do not move together. §7.8 quantifies the second.

§1 Key takeaways

1. A systematic strategy is a rule fixed in advance. That property, not automation or sophistication, makes it testable. It also makes over-testing possible.
2. The rule is a delivery mechanism. The return comes from what the rule holds, so the first question about any strategy is what it is paid for.
3. Every strategy's expected return splits exactly into three parts. The static part is average holdings times average returns. The timing part is the covariance of positions with subsequent returns. The third part is cost.
4. The static part can usually be bought cheaply. Only the timing part needs forecasting skill, and it is the part that noise fakes most easily.
5. On US stocks from 1927 to 2025, a 10-month moving-average rule earned about six-sevenths of its return by being invested. Its timing had a t-statistic of 1.2, yet it halved the worst drawdown in every period.
6. A rule can time risk as well as return. Timing risk shows up in the Sharpe ratio and the drawdown, not in the mean decomposition.
7. The value of skill scales with the square root of the number of independent bets. Diversify across assets and strategies before refining a signal.

2. Where returns come from, and why some survive being known

This section develops the second of the chapter's three ideas: **every durable return is a payment for something.** If it is clear what is being paid for, and who pays, it is possible to say whether the return will survive being widely known. If it is not clear, the history of the field says to assume that the return will not survive.

2.1 The arithmetic nobody escapes

Sharpe (1991) made the starting point unanswerable. The market portfolio's return is the value-weighted average of the returns of everyone who holds stocks. So before costs, the average actively managed dollar earns exactly the market return. After costs, it earns less. Every strategy that beats the market is matched, dollar for dollar, by positions that trail it.

That turns the question "which strategies don't get traded away?" into a more useful one: **who is on the other side, and why do they keep being there?** A strategy's excess return is somebody else's shortfall. It persists only as long as that somebody keeps accepting it.

Economics has a precise answer for when they will. Grossman & Stiglitz (1980) showed that prices cannot be perfectly efficient. If prices reflected all information, nobody would pay to gather it, and then prices could not reflect it. In equilibrium, the people who make prices more accurate earn just enough to cover the cost of doing so. Pedersen (2015) calls the result *efficiently inefficient* markets. They are inefficient enough that active investors are compensated for their costs and risks. They are efficient enough that the marginal active investor earns nothing extra.

For an individual, this has a sharp implication. **A durable excess return is compensation for a cost that the investor bears.** The cost may be risk, illiquidity, a constraint that others face and the investor does not, discomfort or effort. The question is never only "is there a return here?" It is also "can this investor bear the cost behind it more cheaply than the next person who wants it?"

2.2 Four sources, ordered by how long they last

Sort the returns that a systematic strategy can earn by who pays them, and four sources emerge. They differ in exactly the dimension that matters here: whether publicity destroys them.

Source	Who is on the other side	Why they stay there	What kills it	Signature in the returns
A. Risk premium	Investors shedding risk they cannot or will not bear	The risk is real, and aversion to losing money in bad times is stable	A change in risk appetite or in the risk itself; *not* knowledge, because the premium cannot be collected without bearing the risk	Losses concentrated in crashes, recessions and liquidity crises; negative skew
B. Structural premium	Agents trading for reasons other than expected return: hedgers, index funds, leverage-constrained or benchmarked institutions, tax-motivated sellers, anyone who needs to trade *now*	Mandates, regulation, tax law and hedging needs change slowly	The constraint loosening, or arbitrage capital growing large enough to absorb the flow	Returns tied to identifiable flows or dates; limited capacity
C. Behavioural premium	Investors making systematic errors: under-reacting to news, extrapolating growth, overpaying for lottery-like payoffs	The errors are human, and exploiting them is risky and uncomfortable, so arbitrage is limited (Shleifer & Vishny, 1997)	Partly publication and cheap implementation; what survives is what is too painful or costly to arbitrage	Long drawdowns, crashes, large tracking error against benchmarks
D. Informational or mechanical inefficiency	Nobody in particular: slow diffusion of public information, or a mechanical pattern	It lasts only while it is unknown or awkward to trade	Being found. Once it is cheap and safe to trade, capital closes it	High Sharpe ratio, little risk, decay after publication

Examples make the categories concrete. The equity premium, the term premium, the variance risk premium and much of carry are risk premia (A). Four examples are structural (B): futures roll yield driven by producers' hedging, the low-beta anomaly attributed to leverage constraints, index reconstitution, and short-term reversal as payment for supplying liquidity. Momentum, and on one reading value, are behavioural (C). The S&P 500 inclusion effect, the pre-FOMC drift and post-earnings drift in large stocks were informational or mechanical (D). As §2.3 and §5.9 show, they are largely gone.

The categories blur. [Contested] Value is claimed by risk-based explanations (Fama & French, 1993) and by behavioural ones (Lakonishok, Shleifer & Vishny, 1994). Low beta is claimed by leverage constraints (Frazzini & Pedersen, 2014) and by benchmarking (Baker, Bradley & Wurgler, 2011). For persistence, the label matters less than one question: **does the return depend on the other side continuing to do something costly, for reasons that publicity does not change?** Risk aversion, regulation and hedging needs pass that test. Ignorance does not.

[Hypothesis] The ordering A, B, C, D by durability is a synthesis made for this chapter, not an established result. It is what the decay evidence below would predict. The view taken here is that it is the right way to weight any strategy under consideration.

2.3 The evidence on decay

The best evidence on what happens to a return once it is known comes from published stock-return anomalies, because their publication dates are known.

- [Fact] McLean & Pontiff (2016) study 97 predictors from academic journals. Their returns are 26% lower out of sample but before publication. That part is attributable to statistical bias in the original studies. Returns are 58% lower after publication. So roughly a third of the original return appears to be traded away by investors who learned of it. Declines are larger for predictors with higher in-sample returns, and for those concentrated in liquid stocks that are cheap to arbitrage. Trading volume and short interest in the affected stocks rise after publication.
- [Fact] Falck, Rej & Thesmar (2022) find that published anomalies, evaluated outside their original samples, deliver about half their in-sample performance. The decay grows with the year of publication: each year, newly published factors decay about five percentage points more. It also grows with measures of overfitting, such as the number of operations needed to compute the signal. The authors are at a quantitative fund.

- [Fact] Chordia, Subrahmanyam & Tong (2014) find anomaly returns attenuated in the era of decimal prices and cheap, fast trading, when arbitrage became cheaper.
- [Contested] Linnainmaa & Roberts (2018) extend accounting-based anomalies back before 1963. They find most of them absent there, which is what data mining would produce.

A live debate concerns whether the original findings were ever real. [Contested] Hou, Xue & Zhang (2020) control for microcap stocks with NYSE size breakpoints and value weighting. On that basis, 65% of 452 anomalies fail a conventional $|t| \geq 1.96$ hurdle, and 82% fail the multiple-testing hurdle of 2.78. Chen & Zimmermann (2022) use the original papers' own methods. They reproduce 98% of the 161 predictors that were clearly significant in the originals. Jensen, Kelly & Pedersen (2023) find that most factors replicate. The factors cluster into 13 themes, and they work out of sample across 93 countries.

These findings are less contradictory than they look, because they measure different things. The original results are mostly real in the data as the authors constructed it (Chen and Zimmermann). Many of them depend on small, illiquid stocks and on equal weighting, which a real portfolio cannot hold at scale (Hou, Xue and Zhang). What survives out of sample is the broad theme, not the particular signal (Jensen, Kelly and Pedersen). For someone implementing a strategy, the second bar is the relevant one, and the third is the relevant unit of analysis.

The broad cross-asset premia look more durable than narrow stock-level anomalies. [Contested] Baltussen, Swinkels & van Vliet (2021) test 24 factor premia across equity indices, bonds, commodities and currencies, with data back to 1800. They find most of them present out of sample, with limited decay. Hurst, Ooi & Pedersen (2017) find trend-following profitable in every decade since 1880. The contest is over the 2010s, which were poor for several of these premia at once.

The Kenneth French data allow a small version of the McLean–Pontiff exercise for the three most famous US equity factors. Each factor is split at the end of the year in which its best-known paper was published. This is a calculation run for this chapter:

Factor	Paper	Before publication	After publication	Change
Size (SMB)	Banz (1981)	3.59% a year, $t = 2.3$ (1927–1981)	0.09%, $t = 0.1$ (1982–2025)	−97%
Value (HML)	Fama & French (1992)	5.27%, $t = 3.4$ (1927–1992)	1.96%, $t = 1.0$ (1993–2025)	−63%
Momentum	Jegadeesh & Titman (1993)	8.84%, $t = 4.5$ (1927–1993)	4.23%, $t = 1.4$ (1994–2025)	−52%

Three caveats keep this honest. First, these three factors were chosen with hindsight, precisely because they became famous, so they are survivors of a much larger search. Second, the "before" periods include decades that the original studies never looked at. Third, a t-statistic of 1.0 or 1.4 over 30-odd years does not show that a premium has gone. It shows only that 30 years cannot tell (§8.2). What the table does show is that the direction and size of the decline match the cross-sectional evidence. Since 2000, the US value and momentum factors have earned 2.4 and 2.1 points a year, with t-statistics of 1.0 and 0.6.

2.4 Persistent, but uncollectable

Some returns survive in the data precisely because they cannot be collected at scale. That category matters, because it is where most claims of an individual investor's edge live.

- **Skill is competed away at the fund level.** Berk & Green (2004) show that even genuinely skilled managers deliver no excess return to their investors in equilibrium. Money flows to managers with good records until diseconomies of scale erase the edge. The manager captures what is left in fees.
- **Costs kill the fast anomalies.** [Fact] Novy-Marx & Velikov (2016) find that anomalies with low turnover, such as value, profitability and size, mostly survive realistic transaction costs. High-turnover anomalies mostly do not. Short-term reversal is the canonical casualty (§5.8).
- **Machine learning finds the illiquid corners.** Avramov, Cheng & Metzker (2023) find that the profits of machine-learning return predictors concentrate in microcaps and in distressed, hard-to-arbitrage stocks. The profits weaken sharply once those stocks are excluded or costs are charged.
- **Institutional costs are lower than academic proxies.** [Contested] Frazzini, Israel & Moskowitz (2018) use 1.7 trillion dollars of live executions. They find real trading costs an order of magnitude below academic estimates, which implies large capacity for the major factors. The authors' firm runs these strategies.

A small trader's market impact is negligible, so capacity limits that bind a fund do not bind the trader. [Hypothesis] That is a real advantage in niches too small for institutions. Three things partly offset it: wider relative spreads in exactly those

niches, borrowing constraints on the short side, and fixed costs that are large relative to a small account. The net effect is strategy-specific, and §7 is mostly about computing it.

2.5 What an individual has that an institution does not

The comparison is less one-sided than either the "retail can't compete" story or the "retail is nimble" story suggests:

Individual's advantage	Why it matters	Individual's disadvantage	Why it matters
Small size	No market impact; capacity-limited niches are open	Taxes	Pensions and endowments pay none; an individual's after-tax return can trail the pre-tax one by several points a year (§7.2)
Long horizon, no redemptions	Can hold through a multi-year drawdown without being forced to sell	Costs in some instruments	Option spreads, small-cap spreads, borrow fees and margin rates are all worse than institutional terms
No benchmark, no career risk	Can hold positions with large tracking error, which institutions avoid	Leverage and shorting limits	Margin rules, retirement-account restrictions, hard-to-borrow stocks
No mandate	Can combine asset classes, hold cash, use futures	Data and operations	Clean historical data costs money; one person is a single point of failure
Tax control	Can choose accounts, lots and the timing of realisations, and harvest losses	Behaviour	A documented tendency to abandon strategies after losses (§7.9)

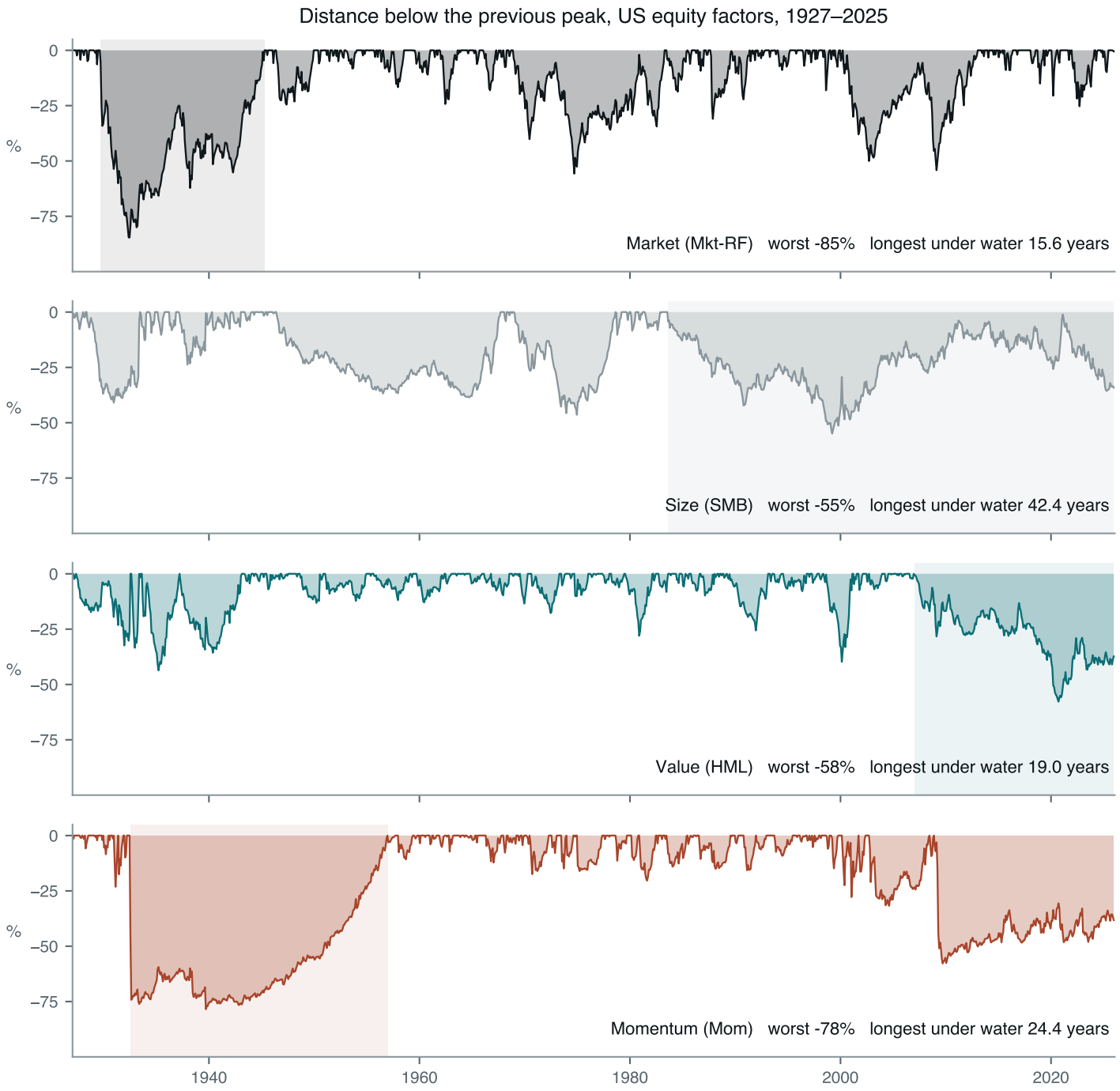
One pattern in this table is worth making explicit. **The institutional constraints that create structural and behavioural premia are exactly the ones that an individual does not face.** These constraints are benchmarks, leverage limits, career risk and redemption risk. So the natural edge for an individual is not a better signal. It is a cheaper tolerance for discomfort. That means holding diversified, well-documented, uncomfortable premia through their drawdowns, cheaply and tax-efficiently. Most individuals demonstrably fail to do this (§7.9). That failure is precisely why it remains an edge for those who can.

2.6 Pain is the price

If a premium could be collected without pain, capital would flow in until it was no longer a premium. So the strategies that survive must contain episodes that make people quit. The pain is not a defect in these strategies. It is the mechanism that keeps them from being arbitraged away. [Hypothesis] That is an argument, not a theorem. The history of the best-known US factors is consistent with it. The figures below come from a calculation run for this chapter, over 1927–2025, with long–short factors compounded as fully collateralised positions:

Factor	Excess return	Sharpe ratio	Worst drawdown
Market	8.26%	0.45	−85%
Size (SMB)	2.04%	0.19	−55%
Value (HML)	4.17%	0.34	−58%
Momentum	7.35%	0.45	−78%
Half value, half momentum	5.76%	0.73	−41%

Factor	Longest below a previous peak	Worst 12 months	10-year windows that lost money
Market	15.6 years (1929–1945)	−66%	14%
Size (SMB)	42.4 years (1983–, not recovered)	−33%	35%
Value (HML)	19.0 years (2007–, not recovered)	−40%	14%
Momentum	24.4 years (1932–1956)	−76%	21%
Half value, half momentum	17.1 years (2008–, not recovered)	−32%	12%



Shaded band: the longest spell below a previous peak. Monthly data from the Kenneth R. French data library.

A value investor who started in January 2007 was still below their high-water mark, in excess-return terms, at the end of 2025. A momentum investor who started in mid-1932 waited 24 years. These are the strategies that *work*. The last row shows the most useful fact in the tables. Value and momentum are negatively correlated ($\rho = -0.41$ here). Asness, Moskowitz & Pedersen (2013) document the same pattern across asset classes. So an even mix of the two had a Sharpe ratio of 0.73, far above either alone, and a shallower worst drawdown. Yet even the mix has spent the 17 years since 2008 below its peak. Diversification changes the size of the pain much more reliably than it changes its duration.

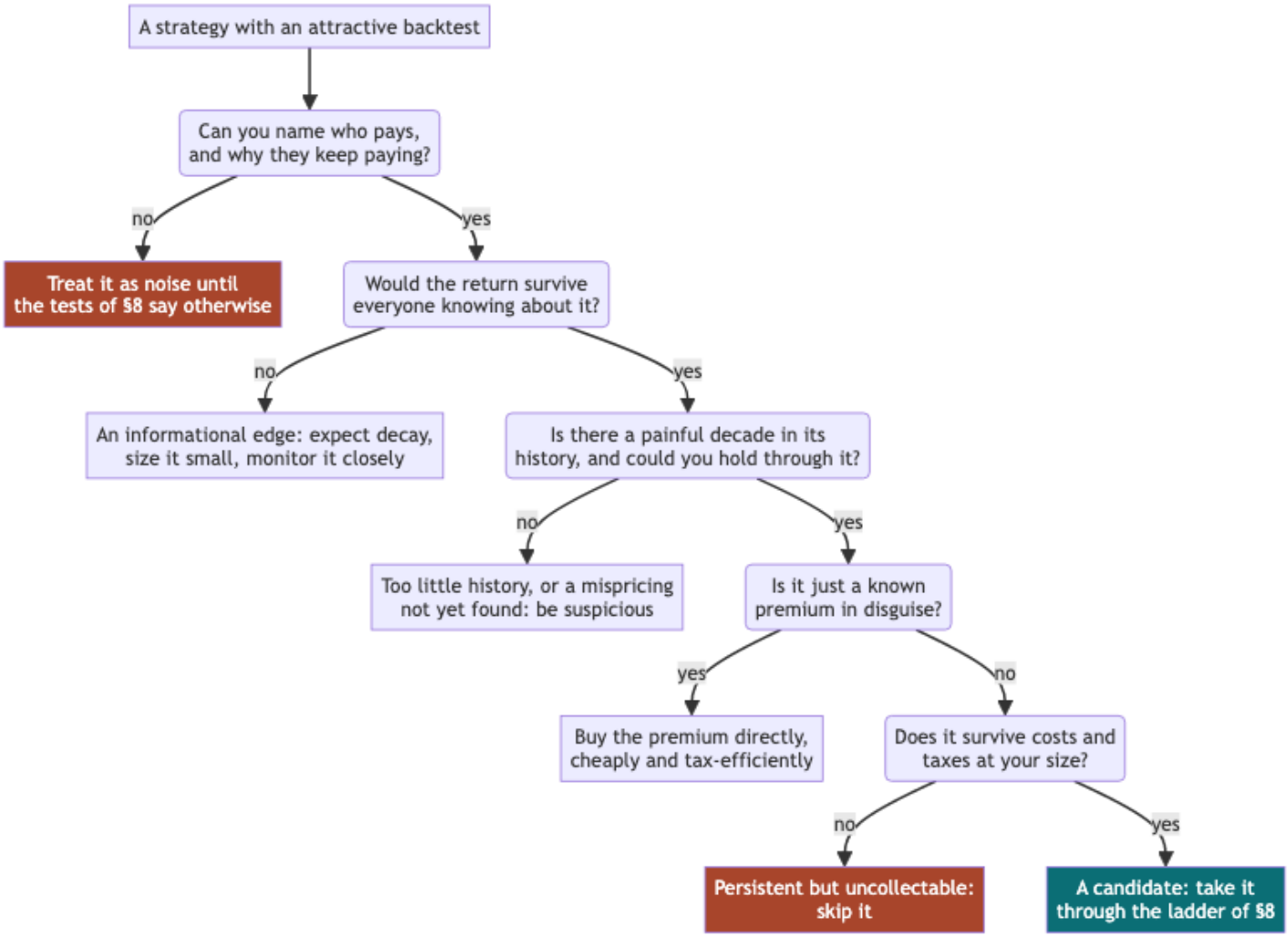
Two things follow for anyone choosing a strategy. First, a backtest with no painful decade in it is not reassuring. Either the sample is too short to contain one, or the return is a mispricing that has not yet been found. Second, the pain that an investor must be able to bear is not the backtest's worst drawdown. That drawdown is one draw from a distribution. What matters is the distribution itself, and §8.8 shows how wide it is, even for a strategy that genuinely works.

2.7 The durability test

This section reduces to a short list of questions to put to any strategy before testing it. They are ordered so that the cheapest to answer come first:

1. **Who is on the other side, and why do they accept a lower expected return?** If they cannot be named, the likeliest answers are “nobody, because the backtest is noise” and “the investor asking the question”.
2. **Would the return survive everyone knowing about it?** Risk premia do, by construction. Informational edges do not.
3. **What does its worst decade look like, and would the investor have held through it?** If the history contains no painful decade, be suspicious, not pleased.
4. **Is it a known premium in disguise?** Split it into static and timing parts (§1.5), and regress it on known factors (§8.6). If it is mostly a premium that can be bought directly, buy it directly.
5. **What does it cost to capture at the investor's size, after tax?** Many persistent returns persist because they cannot be collected.
6. **How easy is it to copy?** A published signal that anyone can compute from free data in an afternoon is the kind whose decay is fastest.

The first five questions work as gates, in order. The sixth does not decide whether to proceed. It indicates only how quickly to expect decay, and therefore how much to allocate. As a flow, with each gate an exit:



§2 Key takeaways

1. Before costs, active management is a zero-sum game against the market portfolio, so every durable excess return has a payer. Ask who pays, and why they keep paying.
2. In equilibrium, a durable return compensates a cost that someone bears: risk, illiquidity, a constraint, discomfort or effort. An investor's edge is bearing that cost more cheaply than the next person.
3. Risk premia and structural premia survive publicity. Behavioural premia partly survive it. Informational edges do not. Weight strategies accordingly.
4. The typical published US stock anomaly lost about 58% of its return after publication. Roughly half of in-sample performance is lost out of sample. The more complex the signal, the faster the decay.
5. Broad cross-asset themes are more durable than individual stock-level signals. The themes are value, momentum, carry, defensive and trend. Many stock-level signals depend on microcaps that cannot be traded at scale.
6. An individual's natural edge is not a better signal but a cheaper tolerance for discomfort: no benchmark, no career risk, no redemptions.
7. Durable premia must hurt. Value spent 19 years below its peak after 2007, and momentum 24 years after 1932. A backtest with no painful decade should be a cause for concern.
8. Value and momentum are negatively correlated. Mixing them raised the Sharpe ratio from 0.34–0.45 to 0.73. Even the mix has been below its 2008 peak since.

3. How the field evolved

The history is short, and it has one recurring shape. Practitioners run rules. Academics test them. The tests find either nothing or a published anomaly. The anomaly then decays, while the rules that encode risk premia survive. The history falls into six eras, each with a one-line thesis.

Rules before theory (1930s–1960s). *Practitioners traded mechanical rules long before anyone could test them.* Richard Donchian's moving-average and channel-breakout rules date from this period. So does one of the first publicly offered managed-futures funds (Futures, Inc., 1949). The academic response was the random walk. Filter rules that bought after a rise of a given percentage earned less than buy-and-hold once commissions were paid (Fama & Blume, 1966). *What lasted:* the rules themselves, which survive almost unchanged in futures trend-following.

Efficiency and its anomalies (1970–1990). *The efficient-markets hypothesis became the null, and the first systematic deviations from the CAPM were catalogued.* Fama (1970) framed efficiency as a testable proposition. The size effect and the price–earnings effect followed. So did long-horizon reversal (De Bondt & Thaler, 1985) and short-term reversal (Jegadeesh, 1990; Lehmann, 1990). In parallel, commodity trading advisers industrialised trend-following in futures. The 1983 "Turtle" experiment showed that a rules-based breakout system could be taught to novices (Faith, 2007). *Limitation:* anomalies were found one at a time, with no accounting for how many had been looked for.

The factor era (1990–2007). *Anomalies were organised into a few factors, and systematic equity investing became an industry.* Fama & French (1992, 1993) reduced the cross-section to market, size and value. Jegadeesh & Titman (1993) added momentum. Carhart (1997) made the four-factor model the standard performance benchmark. Grinold (1989) had already given active management its fundamental law. Brock, Lakonishok & LeBaron (1992) found that simple moving-average rules had predictive power in a century of Dow Jones data. Sullivan, Timmermann & White (1999) showed how much of that survived a correction for the thousands of rules that could have been tried. Lo & MacKinlay (1990b) had already named the problem: data snooping. The collapse of Long-Term Capital Management in 1998 made the limits of arbitrage of Shleifer & Vishny (1997) vivid. *What changed:* "alpha" came to mean mostly "exposure to factors that had not been accounted for".

Crowding and crashes (2007–2012). *Popular systematic strategies turned out to share positions, and to fail together.* In August 2007, quantitative equity funds suffered sudden, simultaneous losses as crowded positions were unwound (Khandani & Lo, 2011). In 2009, momentum crashed when beaten-down stocks rebounded violently (Daniel & Moskowitz, 2016). Trend-followers, by contrast, profited in 2008, which cemented the idea of trend as "crisis alpha". Moskowitz, Ooi & Pedersen (2012) formalised time-series momentum. Asness, Moskowitz & Pedersen (2013) showed value and momentum in every major asset class. *Lasting influence:* the shift from single anomalies to diversified "style premia" across asset classes.

Democratisation and the replication reckoning (2011–2020). *Tools reached individuals just as the literature began to doubt its own results.* Platforms such as Quantopian let thousands of individuals write and backtest algorithms. Quantopian's own study of 888 of them found that backtest Sharpe ratios had almost no power to predict live results (Wiecki, Campbell, Lent & Stauth, 2016). The platform closed in 2020. Bailey, Borwein, López de Prado & Zhu (2014), Harvey, Liu & Zhu (2016) and McLean & Pontiff (2016) quantified overfitting and decay. Hou, Xue & Zhang (2020) found most anomalies fragile. Meanwhile, value endured its longest drawdown, and trend-followers had a lean decade. Short-volatility products imploded in a single day in February 2018 (Augustin, Cheng & Van den Bergen, 2021). For individuals, costs collapsed. US online brokers cut stock commissions to zero in October 2019. Exchange-listed micro futures made diversified futures positions affordable in small accounts.

Retail at scale (2020–). *Individuals became a large, measurable and mostly losing part of short-horizon trading.* Retail participation in single stocks and options surged. The January 2021 episode in GameStop and other heavily shorted stocks made retail order flow a market force. Studies of the retail options boom document large average losses (Bryzgalova, Pavlova & Sikorskaya, 2023; de Silva, Smith & So, 2026). The losses are concentrated in same-day expiries (Beckmeyer, Branger & Gayda, 2023). Machine learning entered asset pricing (Gu, Kelly & Xiu, 2020). Trend and value both had a strong 2022 as interest rates rose. In June 2026, the US$25,000 pattern-day-trader minimum was removed from FINRA's margin rule (FINRA, 2026). [Hypothesis] The newest change is that AI coding assistants have cut the cost of building a research and execution stack. That lowers one barrier to entry. By the logic of §2.3, it also speeds the decay of anything simple.

The milestones, in one table:

Year	Milestone	What changed
1949	Donchian's managed-futures fund	Mechanical trend rules offered to outside investors
1966	Fama & Blume on filter rules	Simple timing rules shown not to beat buy-and-hold after costs
1970	Fama's efficient-markets review	Efficiency became the null hypothesis

Year	Milestone	What changed
1981–1985	Size effect; long-horizon reversal	The first durable-looking deviations from the CAPM
1983	The Turtle experiment	Trend-following shown to be teachable as a rule
1989	The fundamental law	Skill, breadth and implementation linked in one formula
1990	Lo & MacKinlay on data snooping	The multiple-testing problem named
1992–1993	Fama–French factors; momentum	Anomalies organised into factors
1997–1998	Limits of arbitrage; LTCM	Why mispricings can persist, and how arbitrageurs die
2007	The quant quake	Crowding risk in systematic equity made visible
2008–2009	Trend profits; momentum crash	Trend as crisis insurance; momentum's crash risk
2012–2013	Time-series momentum; "everywhere"	Style premia across asset classes
2014–2016	Backtest-overfitting and decay literature	Selection bias measured; the $t > 3$ hurdle
2018	Short-volatility products collapse	The left tail of volatility selling, in one day
2019	Zero commissions; micro futures	Retail trading costs and minimum sizes collapse
2020	Replication studies; ML asset pricing	Most anomalies fragile; broad themes survive
2021–2023	Meme stocks; same-day options	Retail as a force in short-horizon trading, mostly on the losing side
2026	Pattern-day-trader minimum removed	A regulatory barrier to active retail trading gone

§3 Key takeaways

1. Mechanical trading rules are older than the theory used to test them. The rules that survived are mostly the ones that encode a risk premium.
2. Every generation of anomalies has been followed by a reckoning with how many rules were tried. The multiple-testing problem was named in 1990, and it is still the field's central statistical issue.
3. The field moved from single anomalies to factors to diversified style premia. Each step traded headline returns for robustness.
4. Crowding is a risk factor of its own. The 2007 quant quake and the 2009 momentum crash hit precisely the most popular systematic strategies.
5. Tools and costs reached individuals just as the evidence on overfitting matured. The platform that democratised retail backtesting also produced the clearest evidence that its backtests predicted almost nothing.
6. Individuals are now a large part of short-horizon trading, and the evidence says that they lose there on average. The history points instead to low-cost, low-turnover premium harvesting.

4. Foundational references

The references are grouped by kind, because each kind is read differently. Each entry links to a freely readable copy where one exists, such as an author's page, NBER, SSRN or arXiv. Otherwise it links to the publisher, marked *paywalled*. Journal details were checked against Crossref. Where it matters, the body text flags practitioner research from firms that run the strategies in question, at the point of the claim.

4.1 The economics of returns

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4.3 Trend and momentum

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4.5 Volatility and rebalancing

- **Coval, J. D. & Shumway, T. (2001).** "Expected Option Returns." *Journal of Finance* 56(3), 983–1009. [DOI] — Index puts earn strongly negative returns.
- **Bakshi, G. & Kapadia, N. (2003).** "Delta-Hedged Gains and the Negative Market Volatility Risk Premium." *Review of Financial Studies* 16(2), 527–566. [DOI] — The volatility risk premium, isolated.
- **Carr, P. & Wu, L. (2009).** "Variance Risk Premiums." *Review of Financial Studies* 22(3), 1311–1341. [DOI] — Implied against realised variance, measured.
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- **Ilmanen, A. (2012).** "Do Financial Markets Reward Buying or Selling Insurance and Lottery Tickets?" *Financial Analysts Journal* 68(5), 26–36. *Paywalled.* — Selling insurance pays, across markets.
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- **Booth, D. G. & Fama, E. F. (1992).** "Diversification Returns and Asset Contributions." *Financial Analysts Journal* 48(3), 26–32. *Paywalled.* — The diversification return.
- **Willenbrock, S. (2011).** "Diversification Return, Portfolio Rebalancing, and the Commodity Return Puzzle." *Financial Analysts Journal* 67(4), 42–49. [DOI] — What rebalancing does and does not earn.

4.6 Short-horizon, event and calendar effects

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- **Nagel, S. (2012).** "Evaporating Liquidity." *Review of Financial Studies* 25(7), 2005–2039. [DOI] — Reversal returns as the price of liquidity.
- **Gatev, E., Goetzmann, W. N. & Rouwenhorst, K. G. (2006).** "Pairs Trading: Performance of a Relative-Value Arbitrage Rule." *Review of Financial Studies* 19(3), 797–827. [DOI] — The canonical pairs-trading study.
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- **Ball, R. & Brown, P. (1968).** "An Empirical Evaluation of Accounting Income Numbers." *Journal of Accounting Research* 6(2), 159–178. *Paywalled.* — Where post-earnings drift began.
- **Bernard, V. L. & Thomas, J. K. (1989).** "Post-Earnings-Announcement Drift: Delayed Price Response or Risk Premium?" *Journal of Accounting Research* 27 (supplement), 1–36. *Paywalled.*
- **Martineau, C. (2022).** "Rest in Peace Post-Earnings Announcement Drift." *Critical Finance Review* 11(3–4), 613–646. [DOI] — Its disappearance.
- **Shleifer, A. (1986).** "Do Demand Curves for Stocks Slope Down?" *Journal of Finance* 41(3), 579–590. *Paywalled.* — The S&P 500 inclusion effect.
- **Harris, L. & Gurel, E. (1986).** "Price and Volume Effects Associated with Changes in the S&P 500 List: New Evidence for the Existence of Price Pressures." *Journal of Finance* 41(4), 815–829. *Paywalled.*
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- **Keim, D. B. (1983).** "Size-Related Anomalies and Stock Return Seasonality: Further Empirical Evidence." *Journal of Financial Economics* 12(1), 13–32. *Paywalled.*
- **Banz, R. W. (1981).** "The Relationship between Return and Market Value of Common Stocks." *Journal of Financial Economics* 9(1), 3–18. *Paywalled.* — The size effect.
- **Ariel, R. A. (1987).** "A Monthly Effect in Stock Returns." *Journal of Financial Economics* 18(1), 161–174. *Paywalled.*
- **McConnell, J. J. & Xu, W. (2008).** "Equity Returns at the Turn of the Month." *Financial Analysts Journal* 64(2), 49–64. [DOI]
- **Etula, E., Rinne, K., Suominen, M. & Vaittinen, L. (2020).** "Dash for Cash: Monthly Market Impact of Institutional Liquidity Needs." *Review of Financial Studies* 33(1), 75–111. [DOI] — A structural payer for the turn-of-the-month effect.
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- **Gao, L., Han, Y., Li, S. Z. & Zhou, G. (2018).** "Market Intraday Momentum." *Journal of Financial Economics* 129(2), 394–414. [DOI]
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4.7 Evaluation and backtest statistics

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- **Lo, A. W. (2002).** "The Statistics of Sharpe Ratios." *Financial Analysts Journal* 58(4), 36–52. *Paywalled.* — Standard errors and the autocorrelation correction of §8.2.
- **Opdyke, J. D. (2007).** "Comparing Sharpe Ratios: So Where Are the p-Values?" *Journal of Asset Management* 8(5), 308–336. *Paywalled.* — The non-normal standard error.
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- **Bailey, D. H. & López de Prado, M. (2014).** "The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting and Non-Normality." *Journal of Portfolio Management* 40(5), 94–107. — The expected maximum of §8.3.
- **Bailey, D. H., Borwein, J. M., López de Prado, M. & Zhu, Q. J. (2014).** "Pseudo-Mathematics and Financial Charlatanism: The Effects of Backtest Overfitting on Out-of-Sample Performance." *Notices of the American Mathematical Society* 61(5), 458–471. [DOI] — Minimum backtest length.
- **Bailey, D. H., Borwein, J. M., López de Prado, M. & Zhu, Q. J. (2017).** "The Probability of Backtest Overfitting." *Journal of Computational Finance* 20(4). [DOI] — Combinatorially symmetric cross-validation.
- **Harvey, C. R., Liu, Y. & Zhu, H. (2016).** "…and the Cross-Section of Expected Returns." *Review of Financial Studies* 29(1), 5–68. [DOI] — The $t > 3$ hurdle.
- **Harvey, C. R. & Liu, Y. (2015).** "Backtesting." *Journal of Portfolio Management* 42(1), 13–28. [DOI] — Haircutting Sharpe ratios for multiple testing.
- **White, H. (2000).** "A Reality Check for Data Snooping." *Econometrica* 68(5), 1097–1126. *Paywalled.*
- **Hansen, P. R. (2005).** "A Test for Superior Predictive Ability." *Journal of Business & Economic Statistics* 23(4), 365–380. [DOI]
- **Sullivan, R., Timmermann, A. & White, H. (1999).** "Data-Snooping, Technical Trading Rule Performance, and the Bootstrap." *Journal of Finance* 54(5), 1647–1691. [DOI] — The reality check applied to thousands of trading rules.
- **Fama, E. F. & Blume, M. E. (1966).** "Filter Rules and Stock-Market Trading." *Journal of Business* 39(1), 226–241. *Paywalled.* — Simple timing rules fail to beat buy-and-hold after commissions.
- **Brock, W., Lakonishok, J. & LeBaron, B. (1992).** "Simple Technical Trading Rules and the Stochastic Properties of Stock Returns." *Journal of Finance* 47(5), 1731–1764. *Paywalled.*
- **Lo, A. W. & MacKinlay, A. C. (1990b).** "Data-Snooping Biases in Tests of Financial Asset Pricing Models." *Review of Financial Studies* 3(3), 431–467. [DOI]
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- **Suhonen, A., Lennkh, M. & Perez, F. (2017).** "Quantifying Backtest Overfitting in Alternative Beta Strategies." *Journal of Portfolio Management* 43(2), 90–104. [DOI] — A 73% median Sharpe ratio haircut from backtest to live.
- **Arnott, R., Harvey, C. R. & Markowitz, H. (2019).** "A Backtesting Protocol in the Era of Machine Learning." *Journal of Financial Data Science* 1(1), 64–74. [DOI] — A checklist version of §8.
- **Carhart, M. M. (1997).** "On Persistence in Mutual Fund Performance." *Journal of Finance* 52(1), 57–82. *Paywalled.* — The four-factor model used to attribute performance.
- **Magdon-Ismail, M., Atiya, A. F., Pratap, A. & Abu-Mostafa, Y. S. (2004).** "On the Maximum Drawdown of a Brownian Motion." *Journal of Applied Probability* 41(1), 147–161. *Paywalled.* — The distribution of the maximum drawdown.
- **Page, E. S. (1954).** "Continuous Inspection Schemes." *Biometrika* 41(1/2), 100–115. *Paywalled.* — The CUSUM.
- **Shumway, T. (1997).** "The Delisting Bias in CRSP Data." *Journal of Finance* 52(1), 327–340. [DOI] — Missing delisting returns, and their size.

4.8 Implementation and practice

- **Novy-Marx, R. & Velikov, M. (2016).** "A Taxonomy of Anomalies and Their Trading Costs." *Review of Financial Studies* 29(1), 104–147. [DOI] — Which anomalies survive costs.
- **Frazzini, A., Israel, R. & Moskowitz, T. J. (2018).** "Trading Costs." Working paper. — Institutional execution costs from 1.7 trillion dollars of trades. Practitioner authors.
- **Avramov, D., Cheng, S. & Metzker, L. (2023).** "Machine Learning vs. Economic Restrictions: Evidence from Stock Return Predictability." *Management Science* 69(5), 2587–2619. [DOI] — Where machine-learning profits come from.

- **Gu, S., Kelly, B. & Xiu, D. (2020).** "Empirical Asset Pricing via Machine Learning." *Review of Financial Studies* 33(5), 2223–2273. — Machine learning applied to the cross-section of stock returns.
- **Tóth, B., Lempérière, Y., Deremble, C., de Lataillade, J., Kockelkoren, J. & Bouchaud, J.-P. (2011).** "Anomalous Price Impact and the Critical Nature of Liquidity in Financial Markets." *Physical Review X* 1(2), 021006. [DOI] — The square-root law of market impact.
- **Israel, R. & Moskowitz, T. J. (2013).** "The Role of Shorting, Firm Size, and Time on Market Anomalies." *Journal of Financial Economics* 108(2), 275–301. [DOI] — How much of each premium the long side supplies.
- **Blitz, D., Baltussen, G. & van Vliet, P. (2020).** "When Equity Factors Drop Their Shorts." *Financial Analysts Journal* 76(4), 73–99. [DOI] — Long legs, short legs.
- **Israel, R. & Moskowitz, T. J. (2012).** "How Tax Efficient Are Equity Styles?" Working paper. — Momentum and value carry similar tax burdens.
- **Chaudhuri, S. E., Burnham, T. C. & Lo, A. W. (2020).** "An Empirical Evaluation of Tax-Loss-Harvesting Alpha." *Financial Analysts Journal* 76(3), 99–108. [DOI] — About one point a year from harvesting losses.
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- **Avellaneda, M. & Zhang, S. (2010).** "Path-Dependence of Leveraged ETF Returns." *SIAM Journal on Financial Mathematics* 1(1), 586–603. [DOI] — The volatility drag of §7.3.
- **Kelly, J. L. (1956).** "A New Interpretation of Information Rate." *Bell System Technical Journal* 35(4), 917–926. — Growth-optimal betting.
- **Thorp, E. O. (2006).** "The Kelly Criterion in Blackjack, Sports Betting, and the Stock Market." In *Handbook of Asset and Liability Management*, vol. 1, 385–428. *Paywalled.*
- **MacLean, L. C., Thorp, E. O. & Ziemba, W. T. (2011).** "Good and Bad Properties of the Kelly Criterion." In *The Kelly Capital Growth Investment Criterion*, 563–572. *Paywalled.* — Why practitioners bet a fraction.
- **Kaminski, K. M. & Lo, A. W. (2014).** "When Do Stop-Loss Rules Stop Losses?" *Journal of Financial Markets* 18, 234–254. [DOI]
- **Khandani, A. E. & Lo, A. W. (2011).** "What Happened to the Quants in August 2007? Evidence from Factors and Transactions Data." *Journal of Financial Markets* 14(1), 1–46. [DOI] — The quant quake.
- **Harvey, C. R., Rattray, S., Sinclair, A. & Van Hemert, O. (2017).** "Man vs. Machine: Comparing Discretionary and Systematic Hedge Fund Performance." *Journal of Portfolio Management* 43(4), 55–69. [DOI] — Similar risk-adjusted performance, 1996–2014. Practitioner authors.
- **Frazzini, A., Kabiller, D. & Pedersen, L. H. (2018).** "Buffett's Alpha." *Financial Analysts Journal* 74(4), 35–55. [DOI] — A legendary record as a systematic portfolio. Practitioner authors.

4.9 Individual investors

- **Barber, B. M. & Odean, T. (2000).** "Trading Is Hazardous to Your Wealth: The Common Stock Investment Performance of Individual Investors." *Journal of Finance* 55(2), 773–806. [DOI]
- **Barber, B. M., Lee, Y.-T., Liu, Y.-J. & Odean, T. (2009).** "Just How Much Do Individual Investors Lose by Trading?" *Review of Financial Studies* 22(2), 609–632. [DOI]
- **Chague, F., De-Losso, R. & Giovannetti, B. (2019).** "Day Trading for a Living?" Working paper.
- **Dichev, I. D. (2007).** "What Are Stock Investors' Actual Historical Returns? Evidence from Dollar-Weighted Returns." *American Economic Review* 97(1), 386–401. [DOI]
- **Hsu, J., Myers, B. W. & Whitby, R. (2016).** "Timing Poorly: A Guide to Generating Poor Returns While Investing in Successful Strategies." *Journal of Portfolio Management* 42(2), 90–98. [DOI]
- **Bryzgalova, S., Pavlova, A. & Sikorskaya, T. (2023).** "Retail Trading in Options and the Rise of the Big Three Wholesalers." *Journal of Finance* 78(6), 3465–3514. [DOI]
- **de Silva, T., Smith, K. & So, E. C. (2026).** "Losing Is Optional: Retail Option Trading and Expected Announcement Volatility." *Review of Finance* 30(2).
- **Beckmeyer, H., Branger, N. & Gayda, L. (2023).** "Retail Traders Love 0DTE Options… But Should They?" Working paper.

4.10 Critiques and replications

Each of these challenges a result used in this chapter. Each is paired with its claim at the point where the claim is made.

- **Hou, K., Xue, C. & Zhang, L. (2020).** "Replicating Anomalies." *Review of Financial Studies* 33(5), 2019–2133. [DOI] — 65% of 452 anomalies fail once microcaps are controlled for.
- **Linnainmaa, J. T. & Roberts, M. R. (2018).** "The History of the Cross-Section of Stock Returns." *Review of Financial Studies* 31(7), 2606–2649. — Most accounting anomalies are absent before 1963.

- **Kim, A. Y., Tse, Y. & Wald, J. K. (2016).** "Time Series Momentum and Volatility Scaling." *Journal of Financial Markets* 30, 103–124. [DOI] — Trend's returns without the volatility scaling.
- **Huang, D., Li, J., Wang, L. & Zhou, G. (2020).** "Time Series Momentum: Is It There?" *Journal of Financial Economics* 135(3), 774–794. [DOI] — Little asset-by-asset predictability.
- **Korajczyk, R. A. & Sadka, R. (2004).** "Are Momentum Profits Robust to Trading Costs?" *Journal of Finance* 59(3), 1039–1082. [DOI]
- **Lesmond, D. A., Schill, M. J. & Zhou, C. (2004).** "The Illusory Nature of Momentum Profits." *Journal of Financial Economics* 71(2), 349–380. [DOI]
- **Novy-Marx, R. & Velikov, M. (2022).** "Betting Against Betting Against Beta." *Journal of Financial Economics* 143(1), 80–106. [DOI] — How much of betting against beta is construction.
- **Cederburg, S., O'Doherty, M. S., Wang, F. & Yan, X. S. (2020).** "On the Performance of Volatility-Managed Portfolios." *Journal of Financial Economics* 138(1), 95–117. [DOI] — Volatility management fails out of sample across 103 strategies.

4.11 Books

Books without a stable free source are listed without a link.

- **Pedersen, L. H. (2015).** *Efficiently Inefficient: How Smart Money Invests and Market Prices Are Determined.* Princeton University Press. — The best single book on how the major systematic strategies make money, and why.
- **Ilmanen, A. (2011).** *Expected Returns: An Investor's Guide to Harvesting Market Rewards.* Wiley. — The encyclopaedia of return sources. The author is a practitioner.
- **Dimson, E., Marsh, P. & Staunton, M. (2002).** *Triumph of the Optimists: 101 Years of Global Investment Returns.* Princeton University Press. — The long-run evidence on the equity premium.
- **López de Prado, M. (2018).** *Advances in Financial Machine Learning.* Wiley. — Backtest overfitting, purged cross-validation, and much else.
- Grinold, R. C. & Kahn, R. N. (2000). *Active Portfolio Management*, 2nd ed. McGraw-Hill. — The fundamental law in full.
- Carver, R. (2015). *Systematic Trading.* Harriman House. — The most practical book on building and sizing a futures trading system as an individual.
- Chan, E. P. (2021). *Quantitative Trading*, 2nd ed. Wiley. — Infrastructure for an individual's first systematic strategies.
- Antonacci, G. (2014). *Dual Momentum Investing.* McGraw-Hill. — The relative-strength rotation of §7.10.
- Faith, C. (2007). *Way of the Turtle.* McGraw-Hill. — The Turtle experiment and its breakout rules.

4.12 Data, regulation and tax

- **French, K. R.** Data Library. — The factor returns behind §1.6 and §2.
- **FINRA (2026).** "Regulatory Notice 26-10." — The replacement of the pattern-day-trader provisions of Rule 4210.
- **Internal Revenue Service.** Publication 550, *Investment Income and Expenses* — Wash sales, and the treatment of investment income.
- **Internal Revenue Service.** Topic 409, *Capital Gains and Losses*
- **26 U.S.C. section 1256.** Section 1256 contracts marked to market.

4.13 If you only read seven things

1. **Sharpe (1991)** — three pages that frame every question in this chapter.
2. **McLean & Pontiff (2016)** — what happens to a return once it is known.
3. **Pedersen (2015)** — how the durable strategies make money, and why they survive.
4. **Asness, Moskowitz & Pedersen (2013)** — the case for diversified style premia.
5. **Bailey, Borwein, López de Prado & Zhu (2014)** — why the best of many backtests is probably a fluke.
6. **Hurst, Ooi & Pedersen (2017)** — more than a century of evidence for trend-following, and what it does and does not show.
7. **Barber & Odean (2000)** — why the investor, not the strategy, is usually the weakest link.

§4 Key takeaways

1. The foundations are short. Zero-sum arithmetic, the equilibrium price of information and the limits of arbitrage explain which returns can last.

2. The decay and replication literature is the most decision-relevant part of the bibliography. Read it before any single anomaly paper.
3. Much of the evidence on durable strategies comes from firms that run them. The claims are often right, but weigh them with that in mind.
4. The critiques are paired with the claims in the text. Each changes how a strategy should be sized or implemented.

5. The strategy families

This section surveys the strategy families that an individual is likely to meet, judged by the tests of §2. Every family gets the same fields in the same order, so the families can be compared directly:

- **Intuition**: the idea in one or two sentences, before any notation.
- **Rule**: a canonical version, with symbols defined.
- **Who pays**: the source of the return, in the A–D classification of §2.2.
- **Evidence**: what the literature supports, tagged.
- **Failure modes**: how it breaks, with the mechanism. This field carries the most practical value.
- **For an individual**: vehicles, capital, turnover and taxes.
- **Durability**: the verdict taken here, given all of the above.

§5.10 compares the families in one table. Two families have their own chapters in this collection, and their entries here are summaries: trend-following and momentum.

5.1 The benchmark: buy-and-hold, and rebalancing

Intuition. Hold a fixed mix of broad markets, and occasionally trade back to it. It is the simplest systematic strategy there is. Every other strategy in this section has to beat it *after costs and taxes* to be worth running.

Rule. Choose target weights $w^\star$, say 60% stocks and 40% bonds. At each rebalancing date, or whenever a weight drifts outside a band around its target, trade back to $w^\star$. Never rebalancing (pure buy-and-hold) is the other extreme.

Who pays. Risk premia (A): the equity premium and the term premium. Nobody pays the return from rebalancing itself. It is arithmetic, as follows.

A portfolio's compound (geometric) growth rate is approximately its arithmetic mean minus half its variance, $g \approx \mu - \frac{1}{2}\sigma^2$. A portfolio rebalanced to fixed weights w_i therefore grows at about $\sum_i w_i\mu_i - \frac{1}{2}\sigma_p^2$. The weighted average of its constituents' growth rates is $\sum_i w_i\mu_i - \frac{1}{2}\sum_i w_i\sigma_i^2$. Subtracting gives

$$g_p - \sum_i w_i g_i \;\approx\; \tfrac{1}{2}\Big(\sum_i w_i\sigma_i^2 - \sigma_p^2\Big),$$

the **diversification return** (Booth & Fama, 1992; Willenbrock, 2011). Take two uncorrelated assets with 20% volatility, held half and half. They have $\sigma_p^2 = 0.02$ against an average variance of 0.04. So the rebalanced portfolio compounds about one percentage point a year faster than the average of the two assets.

That gain is real but easy to misread. It is measured against the *average* of the constituents. It is not measured against buy-and-hold, which drifts toward whichever asset wins. Whether rebalancing beats buy-and-hold depends on the path. Perold & Sharpe (1988) showed that constant-mix rebalancing has a **concave** payoff. It sells what has risen and buys what has fallen. So it does well when relative prices oscillate, and badly when they trend. Buy-and-hold is linear, and portfolio insurance is convex. Rebalancing is, in effect, a small short position in the volatility of relative performance (§6.3).

Evidence. [Fact] Equities have beaten bills over the long run in every market with a continuous century of data (Dimson, Marsh & Staunton, 2002). In the Kenneth French data, the US market earned 8.3 points a year over T-bills from 1927 to 2025. Its Sharpe ratio was 0.45 and its t-statistic 4.5, in a calculation run for this chapter.

Failure modes. Nothing structural, which is not the same as nothing painful. The US market spent 15.6 years below its 1929 peak in excess-return terms (§2.6). A stock–bond mix depends on the two not falling together. In 2022 both fell as inflation and interest rates rose. The dominant failure is behavioural: selling after a drawdown and buying back after the recovery (§7.9).

For an individual. Hold two to four ETFs, preferably inside a tax-advantaged account. Rebalance annually, or when a weight drifts more than about five percentage points from target. Where possible, rebalance with new contributions, since buying the laggard with fresh cash realises no gains. Costs are close to zero.

Durability. As durable as the risk premia themselves. It is the cheapest and most tax-efficient strategy in this section, and it is the hurdle for all the others.

5.2 Trend-following (time-series momentum)

Intuition. Assets that have risen over the past several months tend to keep rising a little longer. Assets that have fallen tend to keep falling. Hold each market long or short according to its own recent trend. Size each position so that it contributes similar risk, across as many markets as possible.

Rule. The canonical version of Moskowitz, Ooi & Pedersen (2012) is

$$w_{i,t} = \text{sign}\big(r_{i,\,t-12\to t}\big)\,\frac{\sigma^\star}{\widehat{\sigma}_{i,t}},$$

where $r_{i,\,t-12\to t}$ is asset i's excess return over the past 12 months, $\widehat{\sigma}_{i,t}$ is an estimate of its current volatility, and $\sigma^\star$ is a target volatility per position. Moving-average crossovers and channel breakouts are close relatives. The trend-following chapter shows that they are all weighted sums of past returns with different weights. The choice among them matters much less than the horizon.

Who pays. [Contested] Three stories coexist. The behavioural story is that investors under-react to news at first and over-react later. The structural story is that hedgers and central banks move prices slowly and predictably. The insurance story is that trend profits come mostly in large, sustained moves, so the strategy behaves like a long option position. Fung & Hsieh (2001) showed that trend-followers' returns resemble those of lookback straddles, which pay off on big moves in either direction. The insurance reading is the one that matters for durability, because insurance premia do not disappear when understood.

Evidence. [Fact] Moskowitz, Ooi & Pedersen (2012) find significant time-series momentum in 58 futures and forward markets. Hurst, Ooi & Pedersen (2017) extend the evidence to 1880. They find positive average returns in every decade, and good performance in eight of the 10 worst drawdowns of a 60/40 portfolio. The authors' firm runs trend strategies. [Contested] Kim, Tse & Wald (2016) find that much of the performance comes from scaling positions by volatility. Unscaled trend returns look much like buy-and-hold. Huang, Li, Wang & Zhou (2020) find little evidence that each asset's own past return predicts its future return. They attribute the pooled result to differences in average returns across assets. The 2010s were a lean decade for the industry. §1.6 is a microcosm. On a single asset, timing had a t-statistic of 1.2 in 99 years, yet the rule halved the worst drawdown.

Failure modes.

- *Whipsaws.* In choppy, trendless markets the rule buys near tops and sells near bottoms. This steady bleed is the premium paid for the insurance.
- *Sharp reversals.* A V-shaped recovery hurts a rule that de-risked on the way down.
- *Short samples.* A trend system's Sharpe ratio is modest, so a decade of flat returns is well within its normal range (§8.8).
- *Single-market versions.* They lack breadth (§1.7), and they are where most retail implementations live.

For an individual. There are three routes, in increasing order of effort.

1. Apply a moving-average or 12-month momentum filter to five to 10 asset-class ETFs, monthly. The tactical allocation of Faber (2007) and the dual momentum of Antonacci (2014) are well-known versions.
2. Buy a managed-futures fund or ETF, which outsources the operations for a fee.
3. Trade futures directly with micro contracts. This needs a six-figure account for adequate diversification (§7.4) and dependable operations (§7.7).

US futures enjoy favourable tax treatment (§7.2). Switching ETFs in a taxable account generates short-term gains.

Durability. [Contested] Durable. It has a century of evidence and a plausible payer: buyers of crisis insurance. But its Sharpe ratio is modest and its flat periods are long. So it earns its place as a diversifier of equity risk, not as a return engine. The trend-following chapter develops all of this.

5.3 Cross-sectional momentum

Intuition. Among stocks, industries, countries or asset classes, the recent relative winners tend to keep beating the recent relative losers for several months.

Rule. Rank assets by their return over the past 12 months *skipping the most recent month*, $r_{i,\,t-12\to t-1}$. The skip avoids the short-term reversal of §5.8. Buy the top decile or quintile and short the bottom. A long-only version holds the top K. Rebalance monthly.

Who pays. [Contested] Mostly behavioural (C): gradual diffusion of information and under-reaction to news. Part of the return is a premium for crash risk. Momentum suffers sudden, severe losses when markets rebound sharply after a bear market (Daniel & Moskowitz, 2016).

Evidence. [Fact] Jegadeesh & Titman (1993) documented it. It persisted after publication in their own later data (Jegadeesh & Titman, 2001). It appears in most countries and asset classes (Asness, Moskowitz & Pedersen, 2013). A calculation run for this chapter on the French data gives the following. Momentum earned 7.35 points a year from 1927 to 2025, with a Sharpe ratio of 0.45. But it earned 4.23 points after 1993 ($t = 1.4$) and 2.1 points since 2000 ($t = 0.6$). Its worst drawdown was −78% in 1932, and it crashed again in 2009.

[Contested] Whether momentum survives costs is disputed. Korajczyk & Sadka (2004) and Lesmond, Schill & Zhou (2004) argued that trading costs consume much of the profit, which is concentrated in small, expensive-to-trade losers. Frazzini, Israel & Moskowitz (2018) find institutional costs far lower. One counterintuitive result concerns taxes. Israel & Moskowitz (2012) find that momentum and value carry similar tax burdens, despite momentum's turnover being roughly five times higher. Momentum sells its losers and lets its winners run, while value's return comes partly as dividends.

Failure modes.

- *Crashes.* After a bear market, the short leg is full of beaten-down, high-beta stocks that rebound hardest. Scaling positions by recent volatility cushions this. In-sample, it roughly doubles the Sharpe ratio (Barroso & Santa-Clara, 2015). [Contested] Whether that survives out of sample is disputed, as it is for volatility management generally (Cederburg, O'Doherty, Wang & Yan, 2020).
- *Turnover.* The portfolio changes substantially every month.
- *Crowding.* It is among the most widely run systematic strategies.

For an individual. Long-only is the natural implementation. It can be done through a momentum factor ETF, or as a portfolio of 20–50 high-momentum large companies rebalanced monthly or quarterly. The long side captures a good part of the premium. [Fact] Israel & Moskowitz (2013) find that long positions account for about half of momentum's profits. Blitz, Baltussen & van Vliet (2020) find that most of the value added by factor portfolios comes from their long legs.

Durability. [Contested] Durable in the sense of §2: behavioural, painful and global. But it has been weaker in US large stocks since 2000 than its reputation suggests, and it is crash-prone. It belongs in a mix, above all with value (§2.6).

5.4 Value

Intuition. Assets that are cheap relative to a measure of fundamental worth tend to outperform expensive ones over years. For stocks the measure is book value, earnings or cash flow. For commodities and currencies it is long-run reversal or real exchange rates.

Rule. Rank stocks on book-to-market or on a composite of valuation ratios. Buy the cheapest, and short or avoid the dearest. Rebalance annually. Outside equities, Asness, Moskowitz & Pedersen (2013) use the negative of the five-year return as a value measure for commodities and currencies.

Who pays. [Contested] It is either a risk premium (A) or a behavioural premium (C). On the risk reading, cheap firms are distressed and do badly in bad times (Fama & French, 1993). On the behavioural reading, investors extrapolate past growth and overpay for glamour (Lakonishok, Shleifer & Vishny, 1994). Both can be partly true, and for durability they point the same way.

Evidence. [Fact] US value earned 4.2 points a year from 1927 to 2025, with a Sharpe ratio of 0.34, in a calculation run for this chapter. It appears across countries and asset classes (Asness, Moskowitz & Pedersen, 2013). [Contested] After publication it earned 2.0 points a year with $t = 1.0$, and it spent 2007–2020 in the deepest drawdown of its history. Israel, Laursen & Richardson (2021) and Arnott, Harvey, Kalesnik & Linnainmaa (2021) attribute most of that drawdown to cheap stocks getting cheaper relative to expensive ones. That is, the valuation spread widened. They do not attribute it to the premium disappearing. They also note that book value misses intangible assets. Value rebounded in 2021–2022. But in the French data, it had still not regained its 2006 peak by the end of 2025. [Fact] Among the largest stocks the value premium is weak, and the long side supplies about 60% of it (Israel & Moskowitz, 2013).

Failure modes.

- *Droughts* lasting a decade or more, often when interest rates are falling and investors favour long-duration growth stocks.
- *Concentration* in whichever sectors are out of favour.
- *Accounting.* Book value ignores intangible investment, which is now most of what many firms invest in.
- *Value traps.* Some stocks are cheap for a reason.

For an individual. Low turnover and a long holding period make value cheap to run. But its dividends are taxed as they arrive, so it belongs in a tax-advantaged account where possible. Value ETFs are inexpensive. Running value directly requires point-in-time fundamental data (§7.5).

Durability. Durable in the sense of §2, with the longest droughts of any family here. Hold it only in combination, above all with momentum. Hold it only if a decade of underperformance would not force a sale.

5.5 Carry

Intuition. Hold assets that pay more than they cost to finance, on the assumption that prices do not change. Examples are high-yield currencies against low-yield ones, bonds on steep yield curves, commodities whose futures trade below spot, and equity indices with high dividend yields.

Rule. Koijen, Moskowitz, Pedersen & Vrugt (2018) define an asset's carry as its expected return if its price does not change. For a futures contract with spot price S_t and futures price F_t,

$$\text{carry}_t = \frac{S_t - F_t}{F_t},$$

which is the return that a long position earns if the spot price stays where it is and the futures price converges to it. Positive carry means buy. Hold the highest-carry assets long and the lowest short, or trade each asset on the sign of its own carry.

Who pays. A crash-risk premium (A) and hedging pressure (B). Carry trades "go up by the stairs and down by the elevator". They earn steadily, and lose suddenly when risk appetite collapses and leveraged positions are unwound together (Brunnermeier, Nagel & Pedersen, 2008). Currency carry loads on a global risk factor (Lustig, Roussanov & Verdelhan, 2011). In commodities, producers who hedge by selling futures pay speculators to take the other side. The roll return this creates explains much of the long-run return of commodity futures (Gorton & Rouwenhorst, 2006; Erb & Harvey, 2006).

Evidence. [Fact] Carry predicts returns in every major asset class studied, in the cross-section and in the time series (Koijen, Moskowitz, Pedersen & Vrugt, 2018). The authors' firm runs carry strategies. [Fact] Currency carry has pronounced negative skewness (Brunnermeier, Nagel & Pedersen, 2008). [Contested] Crypto has a carry trade of its own: the gap between futures and spot prices. Schmeling, Schrimpf & Todorov (2023) find it large and linked to leveraged, trend-chasing demand, with crash risk to match.

Failure modes.

- *Crashes*, as in currency carry in 1998 and 2008.
- *Crowding.* Carry positions are levered, so unwinds cascade.
- *Concentration.* In commodities, a few contracts supply most of the carry.
- *Counterparty risk.* The crypto basis trade exposes the holder to the exchange itself, as clients of failed exchanges discovered in 2022.

For an individual. Currency forwards are largely closed to retail accounts, but exchange-listed currency futures are not. Bond carry can be expressed with Treasury futures or duration-matched ETFs. Commodity carry can be expressed with futures, or with funds built around roll yield. US futures get the tax treatment of §7.2.

Durability. Durable, because the crash risk that pays it is real. Size it by the crash, not by the steady returns between crashes.

5.6 Defensive: low volatility, low beta, and quality

Intuition. Safer stocks earn more than their risk says they should. Safer here means low beta, low volatility, profitable, stable and conservatively financed. Such stocks often earn about the market's return with noticeably less risk.

Rule. Rank stocks by trailing beta or volatility, or by a quality composite of profitability, growth, safety and payout (Asness, Frazzini & Pedersen, 2019). Hold the safest. The academic "betting against beta" factor (Frazzini & Pedersen, 2014) goes long low-beta stocks levered up to a beta of one. It goes short high-beta stocks levered down to a beta of one.

Who pays. Mostly structural (B). Investors who cannot or will not borrow get more risk by buying high-beta stocks instead. That bids those stocks up and flattens the relation between beta and return (Black, 1972; Frazzini & Pedersen, 2014). Fund managers judged against a benchmark avoid low-volatility stocks, because they create tracking error (Baker, Bradley & Wurgler, 2011). There is a behavioural strand as well (C): a preference for lottery-like stocks with extreme recent returns (Bali, Cakici & Whitelaw, 2011).

Evidence. [Fact] A flat or negative relation between risk and return in the cross-section of stocks was documented early (Haugen & Heins, 1975). High idiosyncratic volatility predicts low returns (Ang, Hodrick, Xing & Zhang, 2006). Profitable firms earn higher returns (Novy-Marx, 2013). [Contested] How much of the betting-against-beta return is collectable is disputed. Novy-Marx & Velikov (2022) show that its construction effectively equal-weights stocks. For each dollar invested, it commits on average 1.05 dollars to stocks in the bottom 1% of market capitalisation. Its non-standard beta estimates also produce part of the result mechanically. Low-turnover defensive strategies do survive costs (Novy-Marx & Velikov, 2016).

Failure modes.

- *Interest-rate sensitivity.* Low-volatility stocks, such as utilities and staples, are bond-like. They suffer when rates rise.
- *Crowding.* Popular low-volatility funds can bid up exactly the stocks they own.
- *Tracking error.* A long-only defensive portfolio lags badly in strong bull markets, which is when investors abandon it.
- *Sector concentration.*

For an individual. Long-only, unlevered versions are easy and cheap. Rank a large-cap universe by one-year volatility and hold the calmest 100, or buy a low-volatility or quality ETF. Turnover is low. Without leverage, the portfolio will usually trail the market in a boom. Its value shows up in drawdowns and in the Sharpe ratio, not in the headline return. The full betting-against-beta construction needs leverage and shorting, which are expensive at retail (§7.4).

Durability. Durable, to the extent that leverage constraints and benchmarking are durable features of institutions, which they are. The magnitude is contested.

5.7 Volatility selling

Intuition. Options cost more, on average, than the volatility that subsequently materialises. Selling them, which is writing insurance, earns the difference in exchange for occasional large losses.

Rule. The **variance risk premium** is the gap between two quantities. One is the market's risk-neutral expectation of future variance, read from option prices. The other is the variance actually realised. The premium has been positive on average for equity indices. There are four common ways to collect it:

- selling one-month index puts against cash (a "put-write");
- writing covered calls;
- selling delta-hedged straddles;
- shorting VIX futures, which usually trade above the spot VIX and drift down toward it.

Who pays. Risk premium (A). Buyers of crash insurance pay for protection against exactly the states in which losses hurt most. These buyers include pension funds, structured-product issuers and anyone hedging a stock portfolio. Index puts earn strongly negative average returns (Coval & Shumway, 2001). Delta-hedged option positions lose money on average (Bakshi & Kapadia, 2003). The variance risk premium is large and negative for the buyer (Carr & Wu, 2009). Retail buyers of short-dated, lottery-like options contribute too (Beckmeyer, Branger & Gayda, 2023; de Silva, Smith & So, 2026).

Evidence. [Fact] Implied variance has exceeded subsequent realised variance on average for major equity indices (Carr & Wu, 2009). [Fact] Across markets, selling insurance and lottery tickets has been rewarded, and buying them penalised (Ilmanen, 2012). Israelov & Nielsen (2015) decompose the covered call into three parts: an equity premium, a volatility premium, and an uncompensated, negative equity-timing component. They argue that most of its risk comes from the uncompensated part. [Fact] On 5 February 2018, a jump in volatility caused a one-day loss of more than 90% in short-volatility exchange-traded products (Augustin, Cheng & Van den Bergen, 2021).

Failure modes.

- *The tail.* The losses are rare and large, and they arrive when everything else in the portfolio is falling. The premium exists because of this, not despite it. A strategy selling one-month out-of-the-money puts can collect small premiums for years, then give back several years of income in weeks.
- *A flattering Sharpe ratio.* Negative skew and fat tails make the Sharpe ratio overstate how attractive the strategy is (§8.5).
- *Margin.* A margin call forces selling at the worst prices, so leverage multiplies the tail.
- *Feedback.* In February 2018, the short-volatility products' own rebalancing required buying VIX futures into the spike.
- *Execution.* Retail sellers of out-of-the-money options give up a large part of the premium to the bid–ask spread.

For an individual. Most US retirement accounts allow cash-secured puts and covered calls. Put spreads cap the tail, at the cost of most of the premium. The popular "wheel" sells puts and, if assigned, sells calls against the shares. It is economically a short-put strategy with discretionary rules (§6.3). Size positions by a stress scenario, such as the one-day fall of more than 20% in October 1987. Never size them by the broker's margin requirement. Tax treatment differs sharply by instrument. Options on broad-based indices, such as the S&P 500 index, are Section 1256 contracts. They are taxed 60% long-term and 40% short-term, regardless of holding period. Options on an ETF that tracks the same index are taxed like stock options (§7.2).

Durability. Durable, because demand for crash insurance is durable. It is also the most dangerous family in this chapter. Its backtests understate its risk, and its Sharpe ratio flatters it. The chapters on implied volatility and dealer hedging cover the machinery.

5.8 Short-horizon reversal and liquidity provision

Intuition. Over days to a month, prices overshoot when someone needs to trade immediately. Buying what has just fallen and selling what has just risen supplies that trader with liquidity. It earns something like a bid–ask spread.

Rule. Weekly or monthly cross-sectional reversal holds each stock in proportion to minus its return relative to the average, $w_{i,t} \propto -(r_{i,t} - \bar{r}_t)$ (Lehmann, 1990; Lo & MacKinlay, 1990a). Pairs trading takes two historically co-moving stocks. When their price ratio diverges, it buys the cheaper and sells the dearer. It unwinds when the ratio converges (Gatev, Goetzmann & Rouwenhorst, 2006). [Practice] Some rules buy a stock index after a short, sharp decline and sell on the rebound. They are popular with retail systematic traders, but they are thinly documented in peer-reviewed work.

Who pays. Structural (B): traders who demand immediacy pay those who supply it. Nagel (2012) shows that reversal returns behave like the returns to liquidity provision. They rise sharply when market volatility rises, and with it the cost of supplying liquidity.

Evidence. [Fact] Short-term reversal is strong before costs (Jegadeesh, 1990; Lehmann, 1990). [Fact] It is among the anomalies that do not survive realistic transaction costs (Novy-Marx & Velikov, 2016). [Fact] Pairs-trading profits have declined over time as statistical-arbitrage capital grew (Do & Faff, 2010).

Failure modes.

- *Costs.* Turnover runs at several hundred percent a year.
- *Catching falling knives.* When liquidity dries up for longer than usual, the strategy keeps buying. The August 2007 quant quake was in part a crowded unwind of such positions (Khandani & Lo, 2011).
- *Competition.* The natural owners of this premium are market makers and high-frequency firms. Their costs are a fraction of anyone else's.

For an individual. Stock-level reversal is generally not viable after retail costs. Index-level rules in liquid ETFs are viable on cost, but under-documented. In a taxable account, their frequent short-term gains are taxed as ordinary income. The June 2026 removal of the pattern-day-trader minimum (§7.4) removes a barrier for small accounts. It does not remove the cost problem.

Durability. The premium is durable, because someone always needs immediacy. But it accrues to the lowest-cost supplier, and an individual is at the wrong end of that cost curve.

5.9 Event, calendar and flow effects: the traded-away gallery

This family is where most "strategies that stopped working" come from. The table groups the best-documented effects by what happened after they became known:

Effect	Original evidence	What happened next	Source (§2.2)
S&P 500 inclusion	Shleifer (1986); Harris & Gurel (1986)	Abnormal return on addition fell from 7.4% in the 1990s to under 1% in the past decade (Greenwood & Sammon, 2025)	D: predictable flow, front-run
Post-earnings drift	Ball & Brown (1968); Bernard & Thomas (1989)	Non-existent in large stocks since 2006 and recently gone in microcaps (Martineau, 2022)	D: slow information processing

Effect	Original evidence	What happened next	Source (§2.2)
Pre-FOMC drift	Lucca & Moench (2015)	Essentially disappeared after 2015 (Kurov, Wolfe & Gilbert, 2021)	D
Weekend effect	French (1980)	Disappeared or reversed after publication (Schwert, 2003)	D
January effect	Rozeff & Kinney (1976); Keim (1983)	Weakened substantially after publication (Schwert, 2003)	B/D: tax-loss selling
Size effect	Banz (1981)	3.6% a year before publication, 0.1% after, in the French data (§2.3)	[Contested]
Turn of the month	Ariel (1987); McConnell & Xu (2008)	Persisted; linked to institutions' month-end liquidity needs (Etula, Rinne, Suominen & Vaittinen, 2020)	B: payment flows [Contested]
"Sell in May"	Bouman & Jacobsen (2002)	Persisted in later data (Andrade, Chhaochharia & Fuerst, 2013)	Unexplained [Contested]
Merger arbitrage	Mitchell & Pulvino (2001)	Persistent; returns resemble selling index puts — small steady gains, losses in market crashes	A: deal-break risk
Intraday momentum	Gao, Han, Li & Zhou (2018)	Linked to option dealers' hedging flows (Baltussen, Da, Lammers & Martens, 2021)	B [Contested]

The pattern is the one that §2 predicts. The effects that were pure information, or predictable flows that were easy to front-run (D), are gone. What remains is tied either to structural flows that someone must execute regardless (B), or to a risk premium (A). Merger arbitrage is the clearest survivor, and the finding of Mitchell & Pulvino explains why. It is paid as insurance, and it loses when insurance pays out.

For an individual, the practical lesson has two parts. Treat any calendar or event strategy as category D until proven otherwise, and expect its backtest to overstate its future. Effects with a structural payer can be worth small allocations. For the rest, decay is not a risk but a base rate.

5.10 The families compared

The table collects the attributes that discriminate among the families. The ratings are the assessment made for this chapter:

Family	Who pays (§2.2)	Durability	Crash risk	Turnover	Feasible for an individual
Buy-and-hold with rebalancing	A	High	Market's own	Very low	Easy (ETFs)
Trend-following	A/B/C	Medium–high [Contested]	Low; positive skew	Medium	ETFs easy; futures need capital
Cross-sectional momentum	C	Medium	High	High	Long-only, moderate
Value	A/C	Medium–high; long droughts	Medium	Low	Long-only, easy
Carry	A/B	High, crash-prone	High	Low–medium	Futures, moderate
Defensive and quality	B/C	Medium–high	Low–medium	Low	Long-only, easy

Family	Who pays (§2.2)	Durability	Crash risk	Turnover	Feasible for an individual
Volatility selling	A	High, crash-prone	Very high	Medium	Easy to do, hard to do safely
Short-term reversal	B	Premium durable, not collectable	Medium	Very high	Poor
Event and calendar	Mostly D	Mostly decayed	Varies	Varies	Poor to moderate

Two structural points stand out. First, the families that an individual can run most easily are the low-turnover ones: buy-and-hold, value, defensive, long-only momentum and ETF-based trend. Low turnover is also where costs and taxes do least damage. Second, the families differ sharply in *when* they lose. Momentum, carry and volatility selling lose in crashes. Trend-following tends to gain in prolonged ones. That difference, not any family's standalone Sharpe ratio, is the main argument for combining them (§7.8).

§5 Key takeaways

1. Buy-and-hold with rebalancing is the benchmark that every other family must beat after costs and taxes. The rebalancing gain is arithmetic. It is worth about one point a year for two uncorrelated 20%-volatility assets, and it is a short bet on trends in relative performance.
2. Trend-following is paid mainly as insurance: small whipsaw losses in ordinary markets, and gains in prolonged crises. Its evidence is long, but its Sharpe ratio is modest, and it needs many markets.
3. Cross-sectional momentum and value are behavioural or risk premia that survived publication in weakened form. Each alone has had multi-decade droughts, and they hedge each other.
4. Carry and volatility selling are genuine risk premia, paid for crash exposure. Size them by the crash, never by the calm.
5. Defensive strategies exploit leverage and benchmark constraints that institutions face and individuals do not. Their long-only versions lag in booms.
6. Short-term reversal is a real premium. It belongs to the lowest-cost liquidity provider, which is not the individual.
7. Calendar and event anomalies without a structural payer have mostly decayed. The S&P 500 inclusion effect fell from 7.4% to under 1%. Post-earnings drift vanished in large stocks after 2006.
8. The families that an individual can run cheaply are the low-turnover ones, and they lose at different times. That is the case for combining them.

6. Taxonomy and equivalences

§5 described nine families as if they were separate things. They are better understood as points in one design space, with a small number of knobs. Several pairs that look different are provably the same trade.

6.1 The master form

Every strategy in §5 can be written as

$$w_{i,t} \;=\; L_t \cdot h\big(\,\psi(\mathcal{I}_{i,t};\,\theta),\;\hat{\sigma}_{i,t},\;\mathcal{C}\,\big),$$

where ψ turns information about asset i into a view. h turns views into positions, given risk estimates $\hat{\sigma}_{i,t}$ and constraints $\mathcal{C}$. L_t sets overall leverage. The slots, and the choices each one takes:

Slot	What it decides	Typical choices
Universe	Which assets i may be held	One index; asset-class ETFs; stocks; futures; options

Slot	What it decides	Typical choices
Information $\mathcal{I}$	What the signal reads	Past prices; company fundamentals; the futures curve; option prices; the calendar; flows
Transformation ψ	How information becomes a view	Time-series (each asset against its own history); cross-sectional (each asset against its peers); spread (one asset against a partner)
Horizon	Look-back and holding period	Days (reversal); months (momentum, trend, carry); years (value)
Sizing h	How views become positions	Equal weight; inverse volatility; an optimiser
Leverage L_t	Overall scale	Fixed; volatility-targeted; drawdown-dependent
Constraints $\mathcal{C}$	What is allowed	Long-only; position caps; turnover limits

The families of §5 are coordinates in this space:

Family	Information	Transformation	Horizon
Trend-following	Past prices	Time-series	1–12 months
Cross-sectional momentum	Past prices	Cross-sectional	12 months, skipping the last
Value	Fundamentals, or long-run prices	Cross-sectional	Years
Carry	Futures curve, interest rates	Either	Months
Defensive	Past volatility and beta; fundamentals	Cross-sectional	Months to years
Volatility selling	Option prices against realised volatility	Time-series	About a month
Short-term reversal	Past prices	Cross-sectional, negative	Days to a month
Calendar and event	The calendar; announcements	Time-series	Days

Written this way, the unexplored combinations become visible: carry within equities, value within futures, trend in single stocks. Several of them are active research areas.

6.2 Which slots matter

Readers consistently spend their effort in the wrong slot. In order of how much they matter:

1. **Information and transformation, because together they decide which premium is being harvested.** They determine the source in §2.2, and therefore durability. This is the choice that deserves the most thought.
2. **Horizon, because it decides turnover, and turnover decides cost and tax.** The same idea at a one-week horizon and at a one-year horizon makes two different businesses.
3. **Sizing, above all scaling by volatility.** Scaling positions inversely to recent volatility is the most reliable single improvement in the risk profile of trend, momentum and diversified portfolios. [Contested] Whether it adds return, and does not just reduce risk, is disputed out of sample (Cederburg, O'Doherty, Wang & Yan, 2020). That it cuts tail risk in equity portfolios is well documented (Harvey, Hoyle, Korgaonkar, Rattray, Sargaison & Van Hemert, 2018).
4. **The functional form of the signal matters least.** Within a family, the choice between a moving-average crossover, an exponentially weighted filter and a raw past return is the choice of a weighting kernel over past returns (Levine & Pedersen, 2016). Performance differences between reasonable choices are small next to the differences between horizons. Backtest optimisation spends most of its time on this slot, which is one reason that optimised backtests disappoint (§8.3).

6.3 The equivalences

Several things that look different are the same trade. Knowing this prevents "diversifying" across three names for one position:

Looks like	Is actually	Status	Why it matters
A covered call (long stock, short call)	A cash-secured short put at the same strike	Exact for European options, by put–call parity	Two ways to hold the same short-volatility, long-equity position; the "wheel" is this trade with assignment
Trend-following	A long position in lookback straddles (Fung & Hsieh, 2001)	Approximate	Positive skew and convexity: it bleeds in quiet markets, like any option buyer
Merger arbitrage	A short position in index puts (Mitchell & Pulvino, 2001)	Approximate	"Market-neutral" hides crash exposure
Constant-mix rebalancing	A concave strategy, short the volatility of relative returns (Perold & Sharpe, 1988)	Exact in the stylised model	Rebalancing and trend-following are opposite bets on the same thing
A stop-loss rule	A momentum bet on the position (Kaminski & Lo, 2014)	Approximate	It helps when returns are positively autocorrelated and costs money under a random walk
Moving-average crossover	A weighted sum of past returns — time-series momentum with a particular kernel (Levine & Pedersen, 2016)	Exact, as a linear filter	Choosing SMA or EMA is choosing weights, not a strategy
Value in commodities or currencies	Five-year price reversal (Asness, Moskowitz & Pedersen, 2013)	By definition	"Value" outside equities is a long-horizon reversal signal
Currency carry	A short position in crash risk (Brunnermeier, Nagel & Pedersen, 2008)	Approximate	Its steady returns are an insurance premium
Dual momentum	Cross-sectional momentum over a few assets plus a time-series filter	Exact, by construction	Two known premia combined, not a third

The first row is worth deriving, because it is exact. Take European options with strike X and time to expiry τ_e, on a stock paying no dividends. Put–call parity says $C - P = S - Xe^{-r_f \tau_e}$, where C and P are the call and put prices, S is the stock price and r_f is the risk-free interest rate. Rearranging,

$$\underbrace{S - C}_{\text{covered call}} \;=\; \underbrace{Xe^{-r_f \tau_e} - P}_{\text{cash plus a short put}},$$

so the two positions have identical payoffs in every state at expiry. With dividends and early exercise, the identity becomes approximate, but not in a way that changes the risk.

One more equivalence connects this section to §1.5. Lo & MacKinlay (1990a) decompose the expected profit of a cross-sectional momentum strategy that holds $w_{i,t} = \frac{1}{N}(r_{i,t} - \bar{r}_t)$, where $\bar{r}_t$ is the equal-weighted average return. Write $\Gamma_{ii} = \mathrm{Cov}(r_{i,t}, r_{i,t+1})$ for each asset's own autocovariance, and Γ_{ij} for the cross-autocovariance $\mathrm{Cov}(r_{i,t}, r_{j,t+1})$. Write σ_μ^2 for the cross-sectional variance of the assets' mean returns μ_i. Then

$$\mathsf{E}[\pi_{t+1}] = \frac{N-1}{N^2} \sum_i \Gamma_{ii} \;-\; \frac{1}{N^2} \sum_{i \neq j} \Gamma_{ij} \;+\; \sigma_\mu^2.$$

The first term is time-series momentum: each asset predicting itself. The second is lead–lag between assets. It *reduces* momentum profits when one asset's rise predicts another's. The third term needs no predictability at all. A strategy that buys past winners will, on average, hold assets with higher mean returns, simply because high-mean assets are more often among past winners. That third term is the static component of §1.5 in another guise. It is the basis of the critique by Huang, Li, Wang & Zhou (2020) of pooled evidence for time-series momentum (§5.2).

6.4 Same name, different thing

The converse problem, one word covering several different trades, causes as much confusion:

- **Momentum** can mean an asset's own past return (time-series), its return relative to peers (cross-sectional), revisions to earnings forecasts (earnings momentum), or intraday continuation driven by hedging flows (§5.9). These are correlated, not interchangeable.
- **Mean reversion** can mean negative autocorrelation of returns at short horizons (§5.8), convergence of valuations (value, §5.4), or convergence of a spread between related assets (pairs). It can also mean the tendency of *volatility* to revert to its average. The last is a statement about risk, not returns.
- **Carry** can mean an interest-rate differential in currencies, the roll yield $(S - F)/F$ in futures, roll-down along a yield curve, a dividend yield or a crypto funding rate. Loosely, it can also mean the time decay collected by an option seller. These carry different risks.
- **Value** means book-to-market, earnings yield or a composite in equities. Elsewhere it means long-horizon reversal, real exchange rates or real yields.
- **Risk parity** can mean inverse-volatility weights, equal risk contributions that account for correlations, or a leveraged multi-asset portfolio built on either. The third is a leverage decision as much as a weighting scheme.
- **Alpha** can mean a regression intercept against some set of factors, the timing component δ of §1.5, or simply return above a benchmark. A strategy can have positive alpha in one sense and none in another. §8.6 shows how to tell which.

§6 Key takeaways

1. Every strategy is a choice of universe, information, transformation, horizon, sizing, leverage and constraints. The families of §5 are coordinates in that space.
2. The information and transformation slots decide which premium is harvested, and therefore its durability. They deserve the most thought.
3. Horizon decides turnover, and turnover decides cost and tax. The signal's functional form matters least, and it is where backtest optimisation spends most of its effort.
4. A covered call is exactly a cash-secured short put. Merger arbitrage and currency carry are approximately short crash insurance. Trend-following is approximately long straddles.
5. Rebalancing and trend-following are opposite bets on the persistence of relative moves, which is why they combine well.
6. Cross-sectional momentum profits split exactly into own-autocorrelation, lead–lag between assets, and dispersion in mean returns. The last needs no predictability at all.
7. "Momentum", "mean reversion", "carry", "value", "risk parity" and "alpha" each name several different trades. Ask which one before comparing anything.

7. Running a strategy as an individual

This section covers the decisions that turn a correct idea into something that runs in a brokerage account. The details are for the United States as of September 2026. The principles travel, but the numbers do not.

7.1 Where the effort should go

People who build their first strategy spend most of their time on the signal. [Practice] Ranked by how much damage each can do, the effort belongs here:

1. **The return source** (§2). It decides durability. Nothing downstream can rescue a strategy that is being paid for nothing.
2. **The data.** A survivorship-biased universe or a mis-adjusted price series produces errors larger than any difference between reasonable signals. It produces them silently (§7.5).
3. **Costs and taxes.** The after-tax difference between two otherwise similar strategies routinely exceeds the pre-tax difference between their signals (§7.2).
4. **Sizing and risk.** Volatility scaling, diversification and position limits decide whether the account survives long enough for the premium to show up (§7.8).
5. **Execution and operations.** A strategy that fails to trade on one day in 20, or trades twice, is a different strategy from the one that was tested (§7.7).
6. **The signal.** It comes last, because within a family its functional form matters least (§6.2).

7.2 Accounts and taxes

US federal tax treatment turns on how long a position is held, and on what kind of instrument it is. [Fact] As of 2026:

- **Short-term gains**, on positions held a year or less, are taxed as ordinary income, at federal rates up to 37%.
- **Long-term gains** are taxed at 0%, 15% or 20%, depending on income.
- A **3.8% net investment income tax** applies above income thresholds. Most states add their own tax.
- **Qualified dividends** are taxed at long-term rates. Other dividends are taxed as ordinary income.
- The **wash-sale rule** disallows a loss if a substantially identical security is bought within 30 days before or after the sale. The disallowed loss is added to the new position's cost basis (IRS Publication 550). A strategy that exits and re-enters the same fund within a month runs straight into it.
- **Section 1256 contracts** include regulated futures, options on broad-based indices and certain currency contracts. They are marked to market at year-end, and taxed 60% at long-term and 40% at short-term rates *regardless of holding period* (26 U.S.C. section 1256).
- In **retirement accounts** (IRAs, 401(k)s), trading gains are not taxed as they occur. But borrowing and short selling are generally unavailable, and futures and option permissions are limited and broker-specific.

The effect is easiest to see with one strategy earning 8% a year before tax for 20 years, taxed four ways. The calculation uses the top federal rates: 37% plus 3.8% short-term, and 20% plus 3.8% long-term. It ignores state tax and the offsetting value of losses:

Treatment	After-tax growth of 1	Annualised
Tax-advantaged account	4.66	8.0%
Bought and held, taxed once at the end at long-term rates	3.79	6.9%
All gains realised yearly, taxed 60/40 as Section 1256 futures	2.95	5.6%
All gains realised yearly at short-term rates	2.52	4.7%

The same pre-tax strategy compounds at anywhere from 4.7% to 8.0% a year. Buy-and-hold in a taxable account earns 6.9%. To match it, a strategy that realises everything short-term each year needs 11.6% before tax. That is an edge of about three and a half points a year, larger than the entire premium that most strategies in §5 earn. Three practical consequences follow:

- **Asset location is a strategy decision.** High-turnover strategies, and those paying non-qualified income, belong in tax-advantaged accounts. Buy-and-hold and Section 1256 exposures tolerate taxable accounts best.
- **Futures and broad index options are tax-favoured.** Consider a trend-follower who holds positions for weeks. At the top brackets, they pay a blended rate of about 31% through futures, against about 41% through ETFs.
- **Tax management is a return source of its own.** [Fact] Harvesting losses in a portfolio of the 500 largest US stocks was worth an estimated 1.10% a year from 1926 to 2018, or 0.85% under the wash-sale rule (Chaudhuri, Burnham & Lo, 2020). It is durable in the sense of §2, because nobody is on the other side of the tax code's asymmetries. It is available only to taxable investors.

7.3 Instruments and vehicles

Vehicle	Best for	Watch for
Individual stocks	Cross-sectional equity strategies: value, momentum, defensive	Point-in-time data; 20–50 names for diversification; spreads in small companies
Exchange-traded funds	Asset-class exposure; factor tilts; ETF-based trend rules	Fees; tracking; short-term gains when switching
Leveraged and inverse ETFs	Short holding periods only	Daily rebalancing makes long-run returns path-dependent (below)
Futures, including micro contracts	Trend, carry, diversified macro; capital efficiency; Section 1256 treatment	Contract size (§7.4); rolling; margin calls; overnight gaps; operational load
Listed options	Volatility selling; defined-risk positions; hedges	Spreads; assignment; the tail; tax treatment by underlying

Vehicle	Best for	Watch for
Managed-futures, factor and multi-strategy funds	Outsourcing implementation	Fees; less control; varying fidelity to the academic premium
Crypto assets and perpetual futures	Trend and carry in a young market	Exchange and counterparty risk; 24-hour operations; tax complexity

Leveraged ETFs deserve their warning. Consider a fund that resets to L times the index every day, over a period in which the index has variance rate σ^2. Its log return is approximately

$$\ln \frac{V_T}{V_0} \;\approx\; L \ln \frac{S_T}{S_0} \;-\; \frac{L^2 - L}{2}\,\sigma^2 T$$

before fees and financing (Avellaneda & Zhang, 2010). The second term is a drag that grows with the square of the leverage. It is 4% a year for a 2× fund on a 20%-volatility index, and 27% a year for a 3× fund on a 30%-volatility index. These products are instruments for short holding periods, not for buy-and-hold strategies.

7.4 Capital, leverage and shorting

Margin. Under Regulation T, a stock position can be bought with 50% initial margin. FINRA's rules set maintenance margin at no less than 25%, and brokers often set it higher. Some brokers offer risk-based portfolio margin to larger accounts, with thresholds set by the broker. [Practice] Retail margin loans cost a benchmark rate plus a spread that is far above institutional financing. That makes levered versions of low-return strategies (§5.6) unattractive.

Day trading. [Fact] Since 4 June 2026, FINRA's margin rule no longer carries the 25,000-dollar minimum for "pattern day traders". Intraday margin standards replace it. Brokers may phase in the change until 20 October 2027, so some will still apply the old rule in the meantime (FINRA, 2026).

Shorting. Borrowing a stock costs a fee. The fee is large for exactly the hard-to-borrow stocks that anomaly short legs favour, and the lender can recall the stock. Short selling is generally unavailable in retirement accounts. [Fact] The long side supplies almost all of the size premium, about 60% of value and half of momentum (Israel & Moskowitz, 2013). So long-only versions keep a good part of each premium.

Futures and the minimum account. Futures embed leverage, and micro contracts made small positions possible. But contracts are indivisible, and that sets a minimum account size for diversification. Suppose the target is K markets with equal risk, average pairwise correlation $\bar{\rho}$, and a portfolio volatility target $\sigma^\star$, on an account of A dollars. With equal risk budgets, portfolio variance is $K(1+(K-1)\bar{\rho})$ times one position's variance. So each position may take annual dollar risk

$$b = \frac{\sigma^\star A}{K}\,D, \qquad D = \sqrt{\frac{K}{1+(K-1)\bar{\rho}}},$$

where $D \geq 1$ is the *diversification multiplier*. One contract of market i carries annual dollar risk $m_i P_i \sigma_i$ (multiplier times price times volatility). So holding at least Q contracts of it requires

$$A \;\geq\; \frac{Q\,K\,m_i P_i \sigma_i}{\sigma^\star D}.$$

A micro S&P 500 contract is 5 dollars times the index. At an index level of 6,500 and 16% volatility, it carries about 5,200 dollars of annual risk. Take 10 markets of similar risk with $\bar{\rho} = 0.2$ ($D = 1.9$) and a 20% volatility target. Holding a single contract of each needs about 138,000 dollars. [Practice] Adjusting positions in reasonable steps takes four or so contracts per market, which needs about 550,000. Some markets carry less risk per contract, which helps. But the conclusion is robust: **a diversified futures portfolio needs a six-figure account, even with micro contracts.** Below that, the same exposures are better bought through ETFs or managed-futures funds. Above it, a smaller universe or a higher volatility target trades diversification for feasibility.

7.5 Data

A backtest is only as honest as its data. The errors that matter most are the ones that make results look better:

- **Survivorship.** The universe must include securities that were later delisted, as of the date the strategy would have seen them. Testing on today's index members builds in hindsight. [Fact] Delisting returns are missing for most US stocks delisted for negative reasons. These returns are large and negative: about −30% in the estimate of Shumway (1997). Dropping them flatters any strategy that holds weak stocks.
- **Adjustments.** Returns must include dividends and be adjusted for splits and other corporate actions. Price-only series understate buy-and-hold returns. Mis-adjusted splits create spurious signals.
- **Point in time.** Fundamental data must be what was known on each date, lagged to its publication date. Later restated values must not be used.
- **Futures continuity.** A continuous futures series stitched across contract rolls is not a price anyone traded. Compute returns within each contract and chain them, or use a back-adjusted series built for that purpose. Decide explicitly when the strategy rolls.
- **Timestamps.** Time zones, exchange holidays, half-days, and the difference between a closing auction price and the last trade all move backtests.

[Practice] Free sources are good for some purposes. The Kenneth French data library covers factor returns, and FRED covers rates and macroeconomic series. But free price feeds typically lack delisted securities and contain adjustment errors. Specialist vendors sell survivorship-free end-of-day data for US stocks and futures at retail prices. Point-in-time fundamentals cost more. Before trusting any source, compare it with a second one over a sample, check prices around known splits, and confirm that delisted names are present.

7.6 Research and backtesting tools

[Practice] Tools fall into three groups:

- **Vectorised backtests** in pandas or NumPy, or libraries built on them, compute a whole history of positions and returns as array operations. They are fast and transparent for daily or monthly portfolio rules. A single misplaced shift turns them into look-ahead machines.
- **Event-driven engines** replay the market bar by bar and route simulated orders. Open-source examples include backtrader, Zipline-reloaded, QuantConnect's LEAN and NautilusTrader. They are slower and closer to live trading. They are the natural choice when the same code will later trade.
- **Hosted platforms** range from cloud research environments to screening services for fundamentals and charting platforms with scripting languages. They trade control for convenience, and their data defines what can be tested.

The tool matters much less than the data (§7.5) and the discipline of the test (§8). What matters in any tool is that positions are decided using only information available at the time. Here is a minimal correct daily loop, for a signal already computed from data up to each date:

```
import numpy as np
import pandas as pd


def backtest(prices: pd.DataFrame, signal: pd.DataFrame,
             vol_target: float = 0.10, cost_bps: float = 5.0) -> pd.Series:
    rets = prices.pct_change()
    vol = rets.ewm(span=60).std() * np.sqrt(252)
    # Decided with data up to and including day t...
    target = signal * (vol_target / vol) / prices.shape[1]
    # ...and held over day t+1. Dropping this shift is the classic look-ahead bug.
    held = target.shift(1)
    turnover = held.diff().abs().sum(axis=1)
    return (held * rets).sum(axis=1) - turnover * cost_bps / 1e4
```

Even this is optimistic in one respect. It assumes that trades happen at the same closing price from which the signal was computed. In practice there are two honest options. One is to compute the signal from prices a few minutes before the close, and trade in the closing auction. The other is to trade at the next day's open, and measure returns from open to open. Either way, test the version that will actually run.

7.7 Execution and operations

Brokers and interfaces. [Practice] Several US brokers offer programmatic access. The range runs from APIs built around a desktop trading application, which needs periodic restarts and re-authentication, to API-first brokers designed for automated clients. Most offer paper-trading environments. Paper fills are generous. They typically assume a trade at the quoted price, without queueing or market impact. So treat paper trading as a test of the plumbing, not of the strategy's costs.

Order types. For daily and slower strategies, market-on-close and limit-on-close orders participate in the exchanges' closing auctions at the official closing price. That is the price most backtests assume. These orders must be entered shortly before the close. Avoid market orders in the first minutes of trading, when spreads are widest. Prefer marketable limit orders, which cap the price paid, to plain market orders.

Execution quality is a measurable cost. [Fact] Commissions on US stocks are mostly zero, but execution prices are not. Schwarz, Barber, Huang, Jorion & Odean (2025) placed 85,000 simultaneous market orders through six accounts at five brokers. They found average round-trip costs ranging from 0.07% to 0.46% across accounts. The costs were driven by the prices that wholesalers gave each broker, and were unrelated to payment for order flow. Record every fill against the price the backtest assumed. That gap, the *implementation shortfall*, is the only cost model that is honest about the investor's own account.

Operations. [Practice] The failures that end retail systematic strategies are mundane:

- *Reconcile before trading.* The broker's record of positions and cash is the source of truth. Compute orders from the difference between it and the target. Never compute them from what the system believes it holds.
- *Make order submission idempotent*, so that a restart does not send the same orders twice. Handle partial fills and rejections explicitly.
- *Know the calendar*: exchange holidays, half-days, contract rolls, and dividend and split dates.
- *Build a kill switch* that flattens or freezes the book. Add alerts that fire when a scheduled run did not happen.
- *Log everything*: inputs, targets, orders and fills. Then any day can be replayed. Run a paper account in parallel with the live one to detect divergence.

7.8 Sizing and combining strategies

Volatility targeting. Scale each position, and then the whole portfolio, to a target volatility: $w_{i,t} \propto \sigma^\star / \widehat{\sigma}_{i,t}$. This is the simplest robust sizing rule. It equalises risk across instruments whose volatilities differ by a factor of 10 or more. It reduces exposure when volatility rises. Its value lies mainly in the risk profile (§6.2).

The Kelly criterion, and why to use a fraction of it. Take a strategy with expected excess return μ and volatility σ. Leverage L produces a long-run growth rate of approximately $g(L) = L\mu - \frac{1}{2}L^2\sigma^2$. It is maximised at $L^\star = \mu/\sigma^2$, where $g(L^\star) = \frac{1}{2}\mathsf{SR}^2$ (Kelly, 1956; Thorp, 2006). The numbers are sobering. A strategy with a Sharpe ratio of 0.5 at 10% volatility has a Kelly leverage of 5, a portfolio volatility of 50%, and a growth rate of 12.5% a year. And $g(2L^\star) = 0$. **So if the true Sharpe ratio is half the estimate, full Kelly on the estimate has zero expected growth.** An error of that size is entirely ordinary, as §8.3 and §8.7 show. Half Kelly keeps three-quarters of the growth at half the volatility. [Practice] Practitioners who use Kelly at all use a fraction of it, and size on a haircut Sharpe ratio (MacLean, Thorp & Ziemba, 2011).

Combining strategies is the most reliable improvement available. Take K strategies with equal Sharpe ratios s, equal risk, and pairwise correlation ρ. The combination has

$$\mathsf{SR}_K = s\sqrt{\frac{K}{1+(K-1)\rho}}.$$

Four strategies with Sharpe ratios of 0.4 combine to 0.80 if uncorrelated, 0.70 at a correlation of 0.1, and 0.51 at 0.5. Negative correlation does better still. Value and momentum had Sharpe ratios of 0.34 and 0.45 alone, and a correlation of −0.41. They combined to 0.73 in the US data (§2.6). The families of §5 lose at different times. Trend loses in choppy markets, carry and volatility selling in crashes, and value in growth booms. That difference, not any one family's standalone Sharpe ratio, is the case for holding several. The chapter on portfolio construction covers how to estimate the correlations and turn them into weights without the estimates destroying the result.

Drawdown rules. [Practice] Many individuals cut risk after losses. That is a momentum bet on their own P&L (§6.3). It helps if the strategy's returns are positively autocorrelated, and costs money if they are not (Kaminski & Lo, 2014). The more robust approach is to size the strategy at the outset, so that its *expected* drawdown distribution (§8.8) is one the investor can live with.

7.9 The behavioural layer

The strongest evidence about individual investors concerns behaviour. It is consistent across countries and decades:

- [Fact] Barber & Odean studied 66,465 US households at a discount broker in 1991–1996. The most active traders earned 11.4% a year, while the market earned 17.9% (Barber & Odean, 2000).
- [Fact] Individual investors in Taiwan lost 3.8 percentage points a year in aggregate through trading. That was equivalent to 2.2% of Taiwan's GDP, and almost all of it came through aggressive orders (Barber, Lee, Liu & Odean, 2009).
- [Fact] Of Brazilians who day-traded equity futures for 300 days or more, 97% lost money. Only 0.4% earned more than a bank teller's wage (Chague, De-Losso & Giovannetti, 2019).
- [Fact] Retail option buyers lose 5–9% on average around earnings announcements, and 10–14% around the most volatile ones (de Silva, Smith & So, 2026). In same-day S&P 500 options, retail traders lost an estimated 358,000 dollars a day after daily expiries were introduced in May 2022 (Beckmeyer, Branger & Gayda, 2023).
- [Fact] Investors' dollar-weighted returns trail the buy-and-hold returns of what they invest in. The gap was 1.3 points a year on the NYSE and AMEX over 1926–2002, and 5.3 points on Nasdaq over 1973–2002 (Dichev, 2007). Value-fund investors trailed their own funds by 1.31 points a year over 1991–2013, through badly timed purchases and sales (Hsu, Myers & Whitby, 2016).

For a systematic investor, the relevant failure is the last one. It is abandoning a sound strategy during a drawdown and returning after the recovery, which converts a patience premium into a timing loss. The defences are procedural:

- Before starting, write down the strategy's expected drawdown distribution (§8.8), and the specific evidence that would justify stopping.
- Automate execution, so that each day does not require a decision.
- Review performance on a fixed, infrequent schedule.

[Practice] A useful test: if the strategy would not be held through twice its worst historical drawdown, reduce the size until it would.

7.10 Common designs, with verdicts

The table judges the designs that individuals most often run by the tests of §2 and §8. The verdicts are the assessment made for this chapter:

Design	What it is	Verdict
Rebalanced stock–bond mix	Two to four index ETFs at fixed weights	The benchmark. Hard to beat after tax, and the default for money that must not be lost to experimentation
Moving-average timing	Hold an index or asset-class ETFs above their 10-month average (Faber, 2007)	Buys drawdown protection at a small whipsaw cost; timing return statistically unproven (§1.6); tax-inefficient in taxable accounts
Relative-strength rotation	Hold the strongest of a few ETFs, with a cash filter (Antonacci, 2014)	Two known premia combined over very few assets; low breadth, high tracking error, results path-dependent
Factor ETF tilts	Value, momentum, quality or low-volatility funds, ideally combined	Cheap, low-turnover access to durable premia; expect decade-long relative droughts in any single factor
Managed-futures fund	A diversified trend strategy bought as a fund	Genuine diversification of equity risk, at a fee; long flat periods
Self-run futures trend and carry	Micro and mini futures across markets	The best breadth and tax treatment; needs a six-figure account and dependable operations (§7.4, §7.7)
Covered calls, cash-secured puts, the wheel	Short index or single-stock volatility	One trade in three forms (§6.3); the income is not extra return; size for the crash
Short-term index mean reversion	Buy an index ETF after a sharp short-term fall	Cheap to trade in liquid ETFs, thinly documented, tax-inefficient; treat as an experiment

Design	What it is	Verdict
Stock-screen strategies	Value, momentum or quality screens on individual stocks	Viable with clean point-in-time data and 20 or more names; avoid the smallest stocks

§7 Key takeaways

1. Spend effort on the return source first, then data, then costs and taxes, then sizing, then operations. Spend it on the signal last.
2. Taxes can matter more than the signal. One 8% pre-tax strategy compounds at 4.7% to 8.0% a year, depending on account and holding period. Put high-turnover strategies in tax-advantaged accounts.
3. US futures and broad index options are taxed 60/40, regardless of holding period. Tax-loss harvesting was worth about 0.85–1.10% a year: a durable return with nobody on the other side.
4. Even with micro contracts, a diversified futures portfolio needs roughly 140,000 dollars to hold one contract in each of 10 markets. Below six figures, use ETFs or funds.
5. Survivorship-free, dividend-adjusted, point-in-time data is non-negotiable. Missing delisting returns alone flatter any strategy that holds weak stocks.
6. Execution quality varies by broker, even at zero commission. Measure fills against the backtest's prices.
7. Never size at full Kelly. If the Sharpe ratio is overestimated by a factor of two, which is ordinary, full Kelly has zero expected growth.
8. Combining strategies that lose at different times is the most reliable improvement available. Four uncorrelated strategies with a Sharpe ratio of 0.4 each combine to 0.8.
9. The documented retail failure is behavioural. Pre-commit to the expected drawdown and to the evidence that would justify stopping.

8. Evaluation

This section develops the third of the chapter's three ideas: **the evidence in a backtest is measured by a t-statistic that grows only with the square root of time, is deflated by every variant tried, and is discounted by costs not yet paid.** Most strategies that an individual would run cannot be statistically confirmed in a lifetime. That is not a counsel of despair. It is why evaluation has to lean on the economic prior of §2 and the attribution of §1.5. It is also why the statistics below are mostly tools for *rejecting* strategies.

8.1 The ladder

Evaluation proceeds from cheap tests to expensive ones. Each stage can kill a strategy before the next is paid for:

Stage	Question	Evidence	Stop if
0. Rationale	Who pays, and why?	The durability test (§2.7)	You cannot name the payer
1. Data audit	Is the data fit to test on?	Survivorship, adjustments, point-in-time (§7.5)	Any hindsight in the universe or the data
2. Signal check	Does the signal predict at all?	Information coefficient; returns by signal quintile	No predictive content, or quintiles out of order
3. Gross backtest	Would the rule have made money?	Sharpe ratio; static and timing parts (§1.5)	The return is a static premium that can be bought directly
4. Net backtest	After costs, taxes and constraints?	Net and after-tax returns; turnover (§7.2)	The edge is smaller than its costs
5. Robustness	Does it survive perturbation?	Nearby parameters; subperiods; other markets	It works only at a sharp peak in parameter space
6. Statistics	Is it distinguishable from luck, given what was tried?	t-statistic; deflated Sharpe ratio (§8.3)	It fails once the trials are counted
7. Attribution	Is it a known premium in disguise?	Factor regressions (§8.6)	The intercept disappears

Stage	Question	Evidence	Stop if
8. Out of sample	Does it work on data not used to build it?	Holdout; walk-forward; small live (§8.7)	It degrades beyond the expected haircut
9. Monitoring	Is it still doing what it was built to do?	Slippage; exposures; drawdown against expectations (§8.8)	The process breaks, or losses leave the expected range

Two orderings matter. The economic question comes first, because it is cheap and because no later stage can answer it. Attribution comes *before* the out-of-sample test. A strategy that is a known premium in disguise should be replaced by the premium, not validated.

8.2 How much a backtest can reveal

The Sharpe ratio is estimated with error. Take n independent, normally distributed returns with per-period Sharpe ratio SR_p. Lo (2002) shows that the estimate has variance of about $(1 + \frac{1}{2}\mathrm{SR}_p^2)/n$. For non-normal returns the variance becomes

$$\mathrm{Var}\left(\widehat{\mathrm{SR}}_p\right) \;\approx\; \frac{1 - \gamma_3\,\mathrm{SR}_p + \frac{\gamma_4 - 1}{4}\,\mathrm{SR}_p^2}{n}$$

(Opdyke, 2007; Bailey & López de Prado, 2012). This reduces to Lo's expression for normal returns ($\gamma_3 = 0$, $\gamma_4 = 3$). Negative skewness raises the uncertainty. That is bad news for exactly the strategies whose Sharpe ratios look best (§8.5).

In annual units over T years, the standard error is about $\sqrt{(1 + \frac{1}{2}\mathrm{SR}^2)/T}$. The t-statistic for a true Sharpe ratio SR grows like $\mathrm{SR}\sqrt{T}$. Setting that equal to a threshold gives the length of history needed:

True Sharpe ratio	Years for expected $t = 2$	Years for expected $t = 3$	Years for 80% power at 5%
0.25	64	144	125
0.5	16	36	31
0.75	7	16	14
1.0	4	9	8
1.5	1.8	4	3.5
2.0	1	2.3	2

The first two columns solve $\mathrm{SR}\sqrt{T} = 2$ or 3. The last column asks for an 80% chance of rejecting "no skill" in a two-sided 5% test. That needs $\mathrm{SR}\sqrt{T} \geq 1.96 + 0.84 = 2.8$.

Read the row for 0.5, which is a good long-run Sharpe ratio for a real premium: **it takes 31 years to have a four-in-five chance of detecting it.** The same arithmetic bounds what an observed number means. A 10-year backtest with an observed Sharpe ratio of 1.0 has a standard error of about $\sqrt{1.5/10} = 0.39$. So its 95% confidence interval runs from roughly 0.25 to 1.75, before any correction for how the strategy was chosen.

Two further traps are common. **Annualising with $\sqrt{12}$ assumes that monthly returns are independent.** Take a first-order monthly autocorrelation of 0.2. The correction of Lo (2002) then gives an annualisation factor of $12/\sqrt{12 + 2 \cdot 11 \cdot 0.2} = 2.96$, not $\sqrt{12} = 3.46$. So the naive figure overstates the Sharpe ratio by 17%. Smoothly marked or illiquid holdings produce exactly this autocorrelation. Second, **a Sharpe ratio computed on a strategy's live track record inherits the same standard error.** Three good years prove as little as three bad ones.

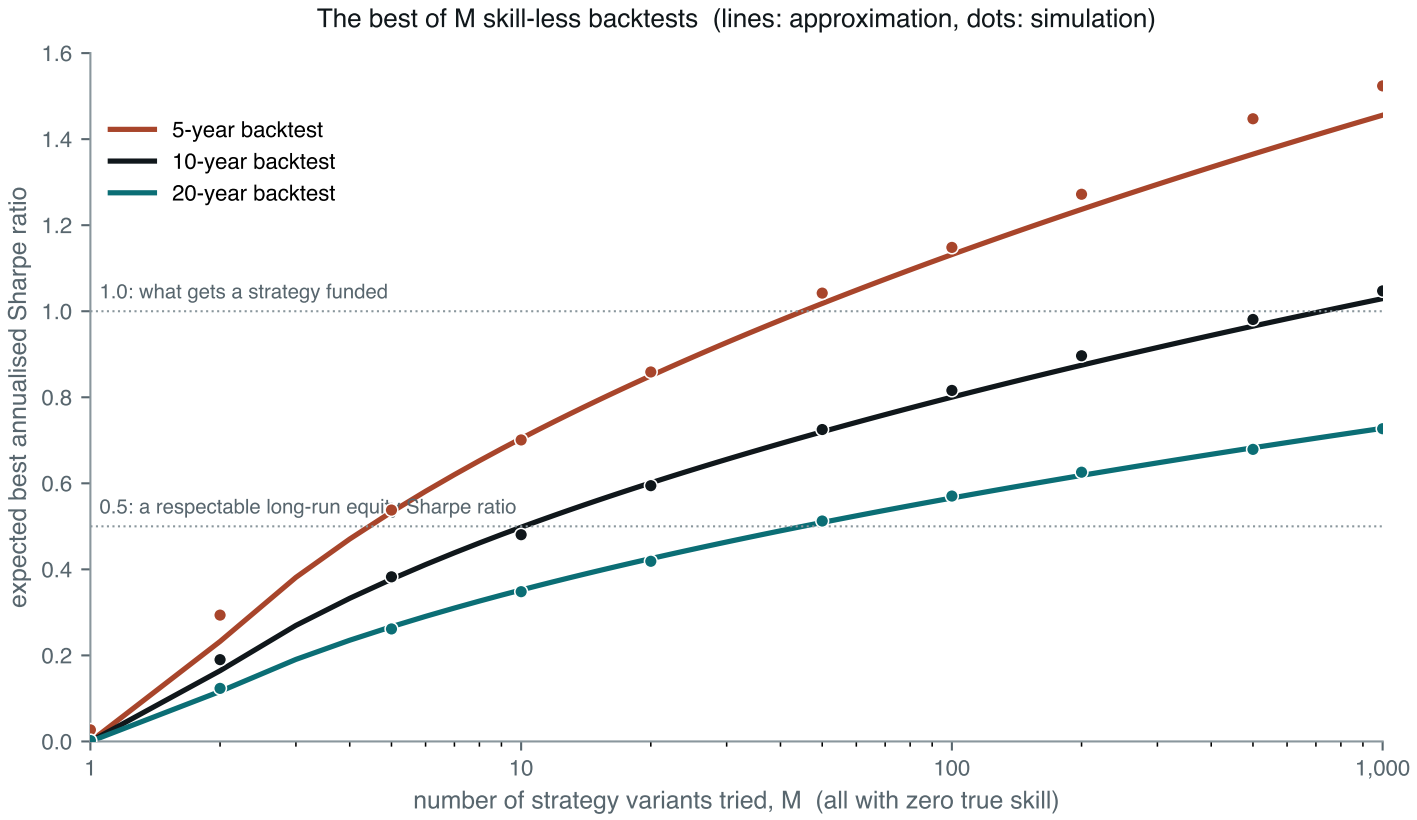
8.3 Selection: the best of many backtests

Every strategy that reaches a presentation is the survivor of many that did not. Suppose M variants are tried, across parameters, universes, filters and periods. None has any skill, and the best is kept. Under that null, each Sharpe ratio estimate over T years has a standard error of about $1/\sqrt{T}$. The expected maximum of M of them is approximately

$$\mathrm{E}\left[\max_{m \leq M} \widehat{\mathrm{SR}}_m\right] \;\approx\; \frac{1}{\sqrt{T}}\left[(1 - \gamma_E)\,\Phi^{-1}\!\left(1 - \tfrac{1}{M}\right) + \gamma_E\,\Phi^{-1}\!\left(1 - \tfrac{1}{Me}\right)\right],$$

where $\gamma_E \approx 0.5772$ is the Euler–Mascheroni constant and e is the base of natural logarithms (Bailey & López de Prado, 2014). The table evaluates it, with a check by simulation, in a calculation run for this chapter:

Variants tried	5-year backtest	10-year backtest	20-year backtest
10	0.70	0.50	0.35
100	1.13	0.80	0.57
1,000	1.46	1.03	0.73



Independent trials. Correlated variants behave like fewer independent ones, which lowers the curve but does not remove it.

One hundred variants on 10 years of data manufacture a Sharpe ratio of 0.8 from nothing. Bailey, Borwein, López de Prado & Zhu (2014) turn this around into a *minimum backtest length*. With five years of data, trying more than about 45 independent configurations makes a Sharpe ratio near 1 the expected result of luck. Correlated variants, such as the same rule with a slightly different look-back, count as fewer independent trials. That lowers the curve, but it does not remove it.

The tools for correcting this, in rough order of effort:

- **Raise the bar.** Harvey, Liu & Zhu (2016) argue that, given the hundreds of factors already tried, a new one should clear $t > 3$, not $t > 2$.
- **Deflate the Sharpe ratio.** The *deflated Sharpe ratio* (Bailey & López de Prado, 2014) is the probability that the true Sharpe ratio is positive. It conditions on the observed ratio, the number and dispersion of the trials, the sample length, and the skewness and kurtosis of returns. It replaces the benchmark of zero with the expected maximum above.
- **Test the whole search.** The reality check of White (2000) and the test of superior predictive ability of Hansen (2005) bootstrap the distribution of the best performance across *all* the rules tried. The reality check was applied to thousands of technical trading rules on a century of Dow Jones data. The best rules survived the correction in-sample, but not in the following decade (Sullivan, Timmermann & White, 1999).
- **Measure overfitting directly.** Split the history into blocks. For every way of choosing half the blocks as a training set, pick the best configuration on the training half, and rank it on the other half. The fraction of splits in which the in-sample winner ranks below the median out of sample estimates the *probability of backtest overfitting* (Bailey, Borwein, López de Prado & Zhu, 2017).

All of these need an honest count of M, and the count is always larger than it feels. [Practice] **Recommendation: keep a trial log from the first day.** Count every variant that was looked at, not every one that was "seriously considered". Decide the

test and the parameter set before looking at results. Hold back the most recent fifth to third of the history, untouched, until the end. It can be used only once.

8.4 The bias catalogue

Beyond selection, these are the recurring ways in which a backtest flatters a strategy:

Bias	What happens	Typical effect	Defence
Look-ahead	Using information not available at the decision time: restated fundamentals, the same bar's close	Can create any Sharpe ratio	Shift positions (§7.6); point-in-time data; timestamp audits
Survivorship	Testing on today's survivors	Overstates returns, most for strategies holding weak stocks	Include delisted securities and historical index membership
Cost omission	Ignoring spread, impact, borrow fees and financing	Largest for high turnover	Charge κu per year; measure real fills
Tax omission	Pre-tax results for a taxable investor	One to three or more points a year (§7.2)	Backtest after tax
Fill optimism	Assuming a limit order fills whenever the price touches it	Fills arrive when the trade is wrong	Assume fills only when the price trades through
Untradeable prices	Stale, illiquid or after-hours prices	Inflates reversal and small-company strategies	Volume filters; tradeable quotes
Short availability	Assuming any stock can be shorted, free	Inflates short legs	Borrow fees; hard-to-borrow filters
Period choice	Starting after a crash, or within one regime	Either direction	Use all available history; report subperiods
Parameter peaks	Choosing the best point on a jagged surface	Inflated and fragile	Prefer plateaus and fewer parameters

8.5 Metrics, and the question each answers

Every metric answers one question and misleads on others:

Metric	Question it answers	How it misleads
Arithmetic mean return	Expected profit per year	It is not what compounds
Compound (geometric) return	What wealth actually did	One number from one path
Volatility	The typical size of moves	Understates tail risk in skewed strategies
Sharpe ratio	Return per unit of risk, comparable across leverage	Flatters negatively skewed strategies; wide standard error (§8.2)
Maximum drawdown	The worst peak-to-trough loss	One draw from a wide distribution, and it grows with sample length (§8.8)
Time under water	How long until a new high	The metric that makes people quit
Skewness and worst month	The shape of the tail	Noisy in small samples
Turnover	Exposure to costs and taxes	—
Hit rate and payoff ratio	How the edge is realised	A 40% hit rate can be excellent; meaningless alone
Correlation with equities in their worst months	Whether it diversifies when needed	Unconditional correlation hides crisis behaviour
Alpha and its t-statistic	Whether it is a known premium (§8.6)	Depends entirely on the factors included

The Sharpe ratio's weakness with skewed strategies deserves a worked example. Take a strategy that earns 1% a month, with a monthly standard deviation of 1.5%, in ordinary months. It loses 30% in one month in 60. Its true mean is about 0.48% a month, and its true standard deviation about 4.2%. That is an annual Sharpe ratio of 0.40. But a four-year backtest misses

the loss month with probability $(59/60)^{48} \approx 45\%$. When it does, the measured Sharpe ratio is $\sqrt{12} \times 1/1.5 \approx 2.3$. So nearly half of all four-year histories of this strategy would show a Sharpe ratio of 2.3. Short-volatility strategies (§5.7) have exactly this shape. That is why they must be judged by stress tests and by skew-aware statistics, not by the Sharpe ratio.

8.6 Attribution: is it a known premium in disguise?

A strategy with a genuine edge should not be reproducible from premia that can be bought cheaply. There are two tests, in order:

1. **Split static from timing** (§1.5). Compare the strategy with a constant position equal to its average exposure, as §1.6 did for the moving-average rule. If the static part explains most of the return, the strategy is an expensive version of a cheap premium.
2. **Regress on known factors.** Estimate

$$\pi_t = \alpha + \sum_k \beta_k F_{k,t} + \varepsilon_t,$$

   where the $F_{k,t}$ are returns of factors that could be held directly. Examples are the market, size, value, momentum and profitability, a trend factor, a carry factor, and the return of a put-writing index. Report α with standard errors that allow for autocorrelation. If α is insignificant, the strategy is a combination of available premia. That is not necessarily bad, but it should be bought as such.

Linear regressions miss option-like exposures. A strategy that is short volatility will show a positive intercept against linear factors. Its losses are concentrated in the rare months that the regression barely weighs. Fung & Hsieh (2001) built option-based factors for exactly this reason. At a minimum, estimate separate betas for up and down markets, or examine the strategy's returns in the market's worst months. [Contested] Frazzini, Kabiller & Pedersen (2018) show how far attribution can go. Leveraged exposure to low-risk and high-quality stocks explains much of Berkshire Hathaway's long-run alpha. The authors' firm runs such strategies.

8.7 Out of sample, on paper, and live

Out-of-sample evidence comes in three strengths:

- **A holdout** is the most recent part of the history, left untouched until the end. It is the cleanest test, and it can be used once.
- **Walk-forward testing** re-fits the strategy on a rolling or expanding window, tests it on the next period, and stitches the test periods together. Choosing among walk-forward methods by their results reintroduces selection. For machine-learning strategies with overlapping labels, purged and embargoed cross-validation (López de Prado, 2018) prevents leakage between training and test sets. The chapter on market regimes covers it.
- **Paper trading and small live trading** test different things. Paper trading tests the plumbing. Only live trading tests costs, and only real money tests the investor's behaviour.

How much should be expected to vanish between backtest and reality? The evidence is consistent:

Study	Setting	Finding
Wiecki, Campbell, Lent & Stauth (2016)	888 algorithms from a retail platform	Backtest Sharpe ratios explained less than 2.5% of the variation in live Sharpe ratios; more backtesting went with a larger gap
Suhonen, Lennkh & Perez (2017)	215 "alternative beta" strategies marketed by investment banks	Median 73% deterioration in Sharpe ratio from backtest to live, over 30 points worse for the most complex
McLean & Pontiff (2016)	97 published anomalies	Returns 26% lower out of sample, 58% lower after publication
Falck, Rej & Thesmar (2022)	Published factors	About half of in-sample performance retained; complexity predicts decay

Harvey & Liu (2015) add that the right haircut is not a constant. Marginal results should be cut by far more than 50%, and very strong ones by less. [Practice] **Recommendation: as a planning figure, assume half of a backtest's Sharpe ratio for a simple, economically motivated strategy with few parameters.** Assume much less for a complex or heavily searched one.

8.8 What a working strategy feels like, and when to stop

The hardest evaluation problem is the live one. The strategy is losing money, and the investor must decide whether it is broken or merely unlucky. The answer starts with knowing what a working strategy looks like from the inside. The table simulates 10 years of daily returns at 10% volatility, for strategies whose *true* Sharpe ratios are known. It is a calculation run for this chapter, with normal returns, so real strategies with fat tails do worse:

True Sharpe ratio	Worst drawdown in 10 years: median (1 in 10 worse than)	Longest spell below a peak: median (1 in 10 longer than)	Share of losing years	Chance the decade loses money
0.3	24% (37%)	4.4 years (8.7)	38%	22%
0.5	21% (32%)	3.3 years (6.9)	31%	8%
1.0	16% (23%)	1.8 years (3.6)	16%	0.2%



Dotted lines mark the median and the 90th percentile. Normal daily returns: real strategies have fatter tails, so these understate the pain.

Drawdowns scale roughly with volatility, so at 20% volatility the depths double. A genuinely good strategy with a Sharpe ratio of 0.5 loses money in about one year in three. One time in 10, it spends nearly seven of 10 years below a previous peak. None of those paths is broken.

It follows that **profit and loss alone cannot reveal promptly that a strategy has stopped working.** The CUSUM of Page (1954) is the classical detector for a shift in a mean. It accumulates the months in which returns fall short of a reference value, halfway between the expected return and zero. It raises an alarm when the shortfall exceeds a threshold. Tune it to raise a false alarm on a working strategy with a Sharpe ratio of 0.5 only about once every 19 years. Then it takes on average 6.5 years (median 5.2) to detect that the same strategy's expected return has fallen to zero. One time in 10, it takes more than 12 years. This is a calculation run for this chapter. A faster detector cries wolf about once a decade.

So stopping rules have to rest on more than the P&L:

1. **Stop at once for process failures.** These include data errors, slippage far beyond the cost model, exposures drifting from the design, and a failure to trade as specified. They are observable quickly and do not need statistics.
2. **Stop when the payer disappears.** The reason the return existed (§2.7) may be gone: a constraint lifted, a flow front-run by arbitrageurs, a market restructured. Then stop, even if recent returns are fine. Good performance at that point is luck.
3. **Treat losses as evidence only against a distribution set in advance.** Before going live, write down the drawdown and time-under-water distributions for the Sharpe ratio actually expected, after the haircut of §8.7. Also write down a threshold at which to reduce size and re-examine the strategy, such as a drawdown beyond the 1-in-20 level of that distribution. Decide it before the drawdown, never during it.

§8 Key takeaways

1. Evaluate in stages, from cheap to expensive: rationale, data, signal, gross and net backtests, robustness, statistics, attribution, out of sample and monitoring. Most strategies should die early.
2. A Sharpe ratio's standard error over T years is about $1/\sqrt{T}$. Detecting a true Sharpe ratio of 0.5 with 80% power takes 31 years. An observed 1.0 over 10 years is consistent with anything from 0.25 to 1.75.
3. Selection manufactures Sharpe ratios. The best of 100 skill-less variants on 10 years of data scores about 0.8. Count every variant, raise the bar to $t > 3$, deflate, and keep a holdout to use once.
4. Negative skew makes the Sharpe ratio lie. A strategy with a true Sharpe ratio of 0.40 shows 2.3 in nearly half of all four-year histories.
5. Attribute before validating. If static exposure or known factors explain the return, buy the premia directly.
6. Expect backtest Sharpe ratios to halve in live trading for simple, well-motivated strategies. Expect them to fall much further for complex or heavily searched ones.
7. A working strategy with a Sharpe ratio of 0.5 loses money one year in three, and can spend seven years under water. A well-tuned detector needs about five to six years to notice that it has died. Stop for process failures and vanished payers, not for P&L alone.

9. Synthesis

9.1 The framework on one page

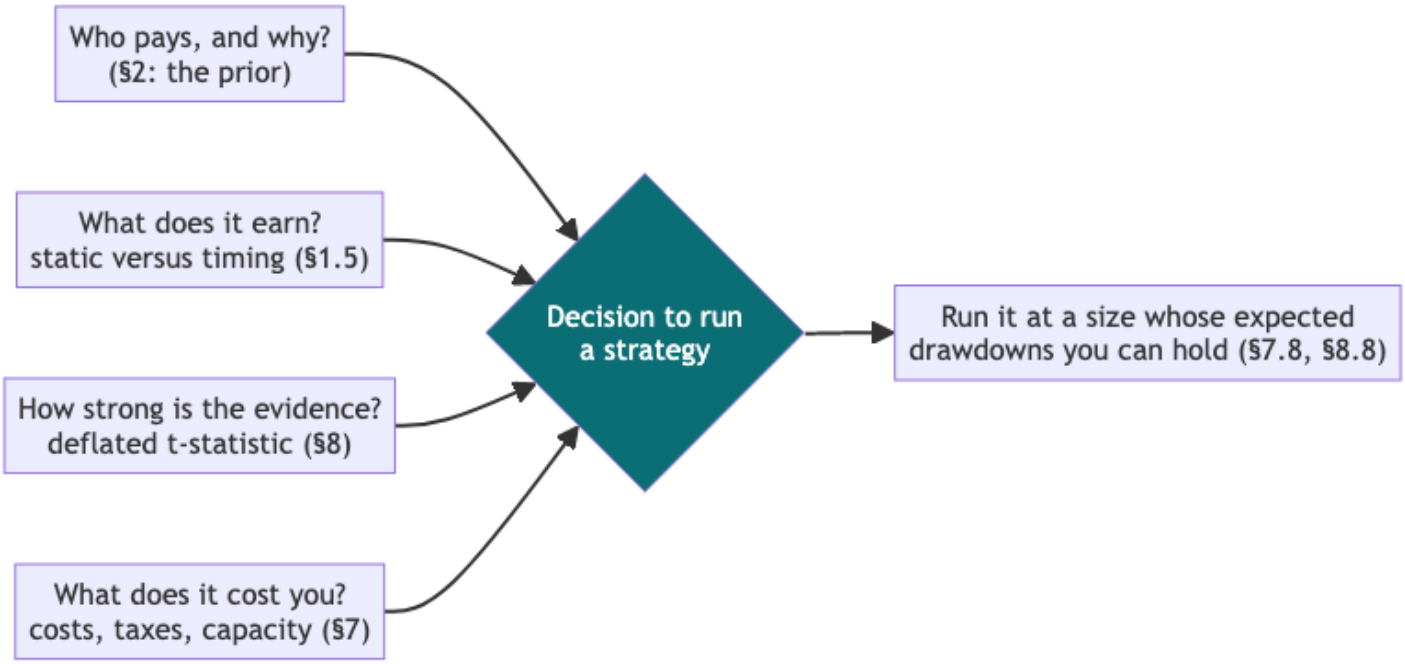
Three ideas generate almost everything in this chapter.

What a strategy earns. Every strategy's expected return splits exactly into three parts (§1.5). The static part is its average holdings times average returns. The timing part is the covariance of its positions with subsequent returns. The third part is cost. The static part can usually be bought cheaply. Only the timing part needs skill, and it is the part that noise fakes.

Why a return lasts. Every durable return is a payment for something, and someone on the other side is paying it (§2). Risk premia and structural premia survive being widely known. Behavioural premia partly survive, protected by the discomfort of holding them. Informational edges do not survive. Durable premia must hurt, and the history of the best of them includes decades under water.

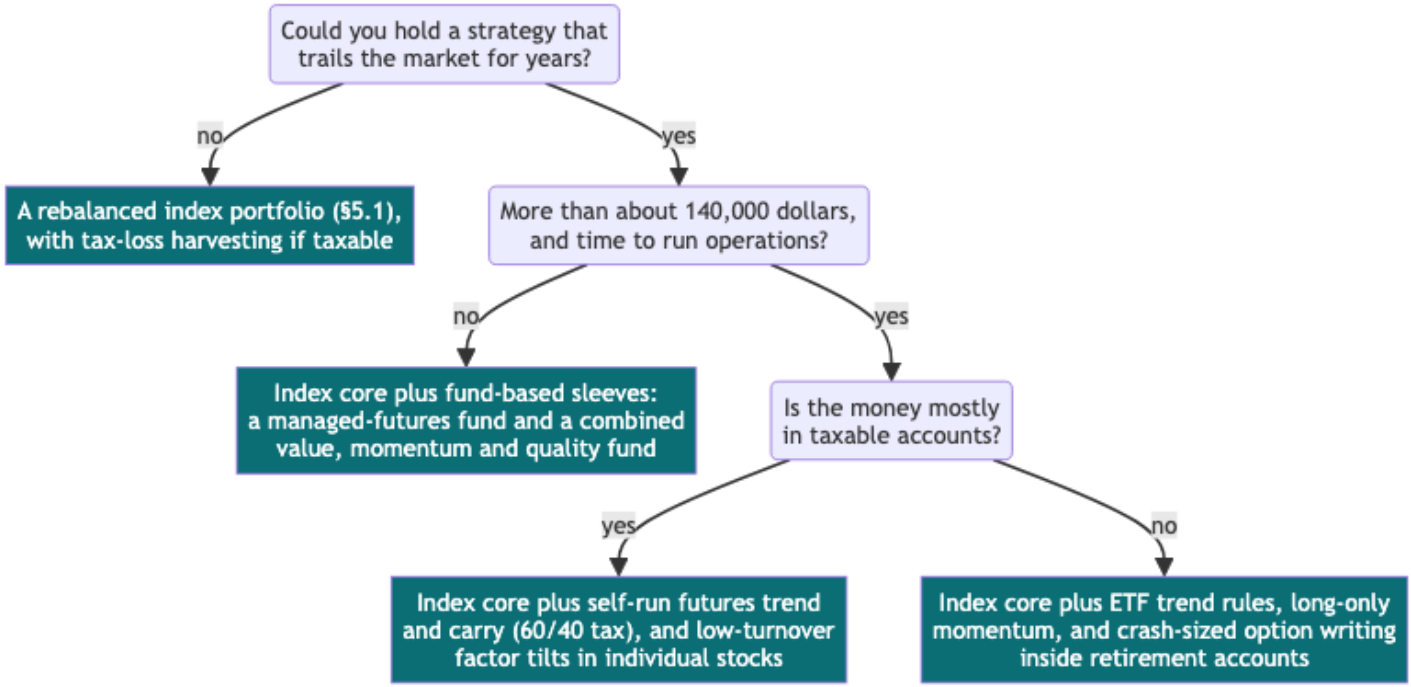
What the evidence can show. A backtest's t-statistic grows only with the square root of time. Every variant tried deflates it, and costs and taxes discount what is left (§8). For most strategies that an individual would run, statistics can reject, but rarely confirm, within a lifetime.

The three ideas interlock. The statistics are weak, so the economic prior has to carry much of the weight: who pays, and why they will keep paying. Much of what looks like skill is static exposure, so the decomposition shows what the statistics are even testing:


Who pays, and why? (§2: the prior)
What does it earn? static versus timing (§1.5)
How strong is the evidence? deflated t-statistic (§8)
What does it cost you? costs, taxes, capacity (§7)
Decision to run a strategy
Run it at a size whose expected drawdowns you can hold (§7.8, §8.8)

9.2 A decision tree for an individual

Three questions change the answer. The first is temperament, the second is capital and time, and the third is tax status:



Whatever the branch, the same disciplines apply:

- name the payer;
- compute the static and timing split;
- model costs and taxes;
- diversify across families that lose at different times;
- write down the expected drawdowns and the stop rules before trading.

An investor who intends to research new signals, instead of harvesting documented premia, should do it with a trial log, a holdout used once, and a small allocation. They should expect whatever they find to decay.

9.3 A staged build, with gates

The early stages are infrastructure, not the interesting part. Skipping them is the usual cause of failure. Each stage has a gate that must be passed before the next begins:

Stage	Build	Gate
0. Infrastructure	Survivorship-free, adjusted data; a backtester with the shift discipline of §7.6; a trial log	Reproduce a published result — the French momentum factor, or §1.6's table — to within tolerance
1. Benchmark	The rebalanced index core, automated, with after-tax accounting	Tracks its index within costs
2. One durable premium	One family from §5 with a named payer: gross and net backtests, the static and timing split, attribution	Positive net, after-tax, deflated evidence, and a return that can be explained
3. Paper, then small live	Run alongside a paper account; record every fill against the model	Three to six months of fills within the cost model, and no missed or duplicated trades
4. Diversify and scale	A second family that loses at different times; sizing by a portfolio volatility target and fractional Kelly on a haircut Sharpe ratio	Realised volatility near target; correlations as expected

Stage	Build	Gate
5. Monitor	Pre-registered drawdown distributions and stop rules; a yearly review of each payer	Stops triggered only by process failures, vanished payers or pre-set thresholds

Stage 0's gate is the most valuable one in the table. A pipeline that cannot reproduce a well-known result from public data cannot be trusted with anything it produces later. Worse, its errors will not be visible.

9.4 Advice for someone starting today

1. **Start from who pays, not from a backtest.** If the payer cannot be named, the backtest shows noise, or the investor's own future losses.
2. **Reproduce a known result before trusting a new one.** It is the only test of the data and code that has a known answer.
3. **Keep a trial log from the first day.** Count every variant, keep a holdout, and use it once.
4. **Split every strategy into static and timing parts.** Most of what looks like timing is exposure that can be bought for less.
5. **Model costs and taxes before optimising anything.** Location, meaning which account holds which strategy, is worth more than most signal refinements.
6. **Prefer low-turnover, economically motivated premia, and expect them to hurt.** A backtest with no painful decade is a warning sign.
7. **Diversify across families that lose at different times before refining any one of them.**
8. **Plan on half the backtest's Sharpe ratio, and size with a fraction of Kelly on that haircut figure.**
9. **Write down the expected drawdowns and the stop rules before going live.** Stop for process failures and vanished payers, not for losses inside the expected range.
10. **Automate execution, reconcile with the broker every day, and measure fills.** The strategy that runs is the one the operations implement, not the one that was tested.

9.5 What is known, and what is not

Some things are settled. [Fact] Equities have earned a premium over cash over long horizons. Published anomalies lose about half their performance out of sample. Backtests overstate live results, and more searching overstates them more. Individuals who trade actively lose on average, mostly through costs and timing. Costs and taxes can exceed the premia they are charged against. Combining premia that lose at different times improves risk-adjusted returns.

Much is contested. [Contested] The open questions are these:

- whether value, momentum, trend and carry will deliver premia as large as their long histories suggest, after the weak 2010s;
- whether their causes are risk or behaviour;
- how much of the low-risk anomaly can be collected;
- whether volatility timing adds return or only reduces risk;
- how much of the retail move into factor funds will be lost to badly timed buying and selling.

Some things are simply unknown. [Hypothesis] It is unknown whether cheaper research tools, including AI-assisted coding, will accelerate the decay of simple signals. It is unknown whether the removal of the pattern-day-trader rule changes retail short-horizon behaviour. It is unknown whether the growth of factor and managed-futures funds has compressed the premia they harvest.

The view taken here is that, for an individual, the reliable edges are not predictive. They are cost and tax efficiency, diversification across premia with named payers, and the patience to hold those premia through the drawdowns that keep them alive. Better prediction is available too. But it is the edge with the most competition, the fastest decay and the weakest evidence, and it is where individuals most often lose.

§9 Key takeaways

1. What a strategy earns, why the return lasts and what the evidence can show are three separate questions. Answer all three before trading.
2. Weak statistics make the economic prior decisive. The static–timing split shows whether there is any skill to test.

3. For most individuals, the right structure is an index core plus sleeves of durable premia, chosen by temperament, capital and tax status.
4. Build in stages with gates. Treat the ability to reproduce a published result as the first and most important gate.
5. The individual's reliable edges are efficiency, diversification and patience, not prediction.

Appendix A. Concepts and prerequisites

This appendix explains everything that the main text relies on without stopping to explain. It is written for a reader who is mathematically comfortable but does not work in finance. Such a reader meets "excess return", "put–call parity" or "the deflated Sharpe ratio" somewhere in the text. They want the idea, not a dictionary definition, and would rather not leave the chapter to get it.

Entries are ordered by **dependency**, not alphabetically: later entries use earlier ones. They are grouped into five parts, which are themselves in dependency order. Each entry gives the idea in words first, then the formal definition, then why it appears here, then where to go deeper. Notation follows the main text's notation table.

Index. Where each concept first matters:

Concept	First used	Concept	First used
Returns: excess and compound	§1.2	Standard errors, t, power	§1.5
Covariance and autocorrelation	§1.5	Regression and alpha	§1.6
The Sharpe ratio	§1.1	Multiple testing	§1.1
Skewness and kurtosis	§2.2	Probabilistic, deflated Sharpe	§8.3
Drawdowns	§1.6	Bootstrap, reality check	§8.3
Dollar-weighted returns	§7.9	Holdouts, cross-validation	§8.3
Costs and turnover	§1.2	The CUSUM	§8.8
Beta and the CAPM	§2.2	Futures	§1.2
Market efficiency	§2.1	Options, put–call parity	§1.5
Risk premia and arbitrage	§2.2	Convex and concave strategies	§5.1
Benchmarks and tracking error	§2.2	Variance risk premium	§5.7
Long–short factors	§1.4	ETFs, leveraged ETFs	§1.2
The fundamental law	§1.7	Orders and execution	§1.2
Volatility targeting	§1.5	US taxes on trading	§1.6
The Kelly criterion	§7.8	Margin, day-trading rules	§2.5

Part I — Returns and risk. The quantities that every table in the chapter reports.

A.1 Returns: simple, excess, arithmetic and compound

The idea. A return is the proportional change in wealth over a period, counting dividends and coupons. Three distinctions run through the chapter. First, an *excess* return subtracts what cash would have earned. Anyone can earn the cash rate without skill or risk, so what a strategy is paid for is the excess. Second, an *arithmetic* average of yearly returns overstates what wealth actually grows at. Losses hurt more than equal gains help: a rise of 50% followed by a fall of 50% leaves 75% of the starting wealth. Third, the gap between the two averages grows with volatility. That is why it is called *volatility drag*.

Formally. The simple return is $R_t = (P_t + D_t)/P_{t-1} - 1$, for a price P and cash distribution D. The excess return is $r_t = R_t - R_{f,t}$, for the cash rate $R_{f,t}$. The arithmetic mean is $\mu = \mathsf{E}[r]$. The compound growth rate is $g = \mathsf{E}[\ln(1+R)]$. A second-order expansion $\ln(1+x) \approx x - \frac{1}{2}x^2$ gives

$$g \;\approx\; \mu - \tfrac{1}{2}\sigma^2.$$

Why it appears here. The notation table defines r as an excess return. The diversification return of §5.1 is a difference of compound growth rates. The leveraged-ETF drag of §7.3 is the $\frac{1}{2}\sigma^2$ term scaled by leverage. The Kelly growth rate of §7.8 is the same approximation applied to a levered position.

Deeper. The chapter on simple and log returns in this collection.

A.2 Variance, covariance, correlation and autocorrelation

The idea. Variance measures how spread out a quantity is. Covariance measures whether two quantities tend to be high or low together. Correlation rescales covariance to lie between −1 and 1, so that it no longer depends on units. Autocovariance is the covariance of a series with its own past. It asks whether this month's return says anything about next month's.

Formally. $\mathrm{Var}(X) = \mathrm{E}[(X - \mathrm{E}\,X)^2]$, and $\mathrm{Cov}(X, Y) = \mathrm{E}[(X - \mathrm{E}\,X)(Y - \mathrm{E}\,Y)]$. The latter rearranges to the identity used in §1.5,

$$\mathrm{E}[XY] = \mathrm{E}[X]\,\mathrm{E}[Y] + \mathrm{Cov}(X, Y).$$

Correlation is $\rho = \mathrm{Cov}(X, Y)/(\sigma_X \sigma_Y)$. For a return series, the lag-k autocovariance is $\mathrm{Cov}(r_t, r_{t+k})$. The autocorrelation ρ_k divides it by the variance. Between two series, the cross-autocovariance $\mathrm{Cov}(r_{i,t}, r_{j,t+1})$ measures whether one asset's return leads the other's.

Why it appears here. §1.5 applies the identity above term by term. The decomposition of momentum profits in §6.3 is built from own- and cross-autocovariances. The correction to annualisation in §8.2 depends on autocorrelation.

Deeper. Section 2 of Foundations of Econometrics.

A.3 The Sharpe ratio, and annualisation

The idea. The Sharpe ratio is return per unit of risk. Leverage multiplies a strategy's excess return and its volatility by the same factor, so their ratio does not change with leverage. That makes it the natural way to compare strategies that could be scaled up or down. It is a poor guide when volatility does not describe risk well (A.4).

Formally. $\mathrm{SR} = \mu/\sigma$ for excess returns. With q periods a year and independent returns, the mean scales by q and the standard deviation by $\sqrt{q}$. So an annual Sharpe ratio is $\sqrt{q}$ times a per-period one. With autocorrelated returns, the factor is instead $q \Big/ \sqrt{q + 2\sum_{k=1}^{q-1}(q-k)\rho_k}$ (Lo, 2002). This is smaller than $\sqrt{q}$ when the ρ_k are positive. Two simpler descriptive ratios appear in §8.5. The *hit rate* is the share of periods with a gain. The *payoff ratio* is the average gain divided by the average loss. The mean return is positive exactly when the hit rate times the payoff ratio exceeds one minus the hit rate. So a 40% hit rate is profitable whenever gains average more than 1.5 times losses. Neither ratio alone says whether a strategy makes money.

Why it appears here. Almost everywhere. Its sampling error is the subject of §8.2, its behaviour under selection of §8.3, and its failure for skewed strategies of §8.5.

Deeper. Sharpe (1994). Lo (2002).

A.4 Skewness, kurtosis and fat tails

The idea. Volatility describes the typical size of a move, not the shape of the distribution. *Skewness* measures asymmetry. A negatively skewed strategy makes small gains often and suffers large losses rarely, which is what selling insurance looks like. *Kurtosis* measures how much probability sits in the tails. "Fat tails" means that extreme moves are more common than a normal distribution allows.

Formally. With mean μ and standard deviation σ, $\gamma_3 = \mathrm{E}[(r-\mu)^3]/\sigma^3$ and $\gamma_4 = \mathrm{E}[(r-\mu)^4]/\sigma^4$. A normal distribution has $\gamma_3 = 0$ and $\gamma_4 = 3$. *Excess* kurtosis is $\gamma_4 - 3$.

Why it appears here. Carry and volatility selling are negatively skewed (§2.2, §5.5, §5.7). Skewness and kurtosis enter the Sharpe ratio's standard error (§8.2). The worked example of §8.5 shows a negatively skewed strategy whose short-sample Sharpe ratio is almost six times its true value.

Deeper. Opdyke (2007). Bailey & López de Prado (2012).

A.5 Drawdowns and time under water

The idea. A drawdown is how far wealth sits below its previous high. It is the quantity that investors actually feel, and the one that makes them abandon strategies. The maximum drawdown of a history is a single draw from a wide distribution. It grows with the length of the history, because longer histories have more chances to contain a bad spell.

Formally. With wealth W_t, the drawdown is $D_t = W_t / \max_{s \le t} W_s - 1 \le 0$. The maximum drawdown is $\min_t D_t$. Time under water is the length of a spell during which $D_t < 0$.

Why it appears here. §2.6 reports the worst drawdowns and longest spells of the US factors. §8.8 shows their distribution for strategies that work.

Deeper. Magdon-Ismail, Atiya, Pratap & Abu-Mostafa (2004).

A.6 Time-weighted and dollar-weighted returns

The idea. A fund's published return is *time-weighted*. It measures what one dollar invested at the start and left alone would have earned. An investor who adds money after good years and withdraws after bad ones earns something different. That is a *dollar-weighted* return, which counts each period in proportion to how much money was invested during it. The gap between the two is the cost of the investor's own timing.

Formally. The time-weighted return compounds the period returns, $\prod_t (1 + R_t) - 1$. The dollar-weighted return is the internal rate of return y that makes the present value of the investor's cash flows zero, $\sum_t C_t/(1+y)^t = 0$. Contributions count as negative flows. Withdrawals and the final balance count as positive ones.

Why it appears here. In §7.9, Dichev (2007) and Hsu, Myers & Whitby (2016) measure the investor return gap as the difference between the two.

Deeper. Dichev (2007).

A.7 Basis points, turnover and the anatomy of trading costs

The idea. Trading costs are small per trade and large per year. So they are measured in basis points, hundredths of a percent, and charged against turnover. They come from several sources:

- commissions;
- crossing half the gap between the best bid and offer;
- moving the price against oneself when trading size (market impact);
- borrowing stock to short it;
- financing leverage.

Formally. One basis point is 0.01%. One-way turnover u is the sum of $|\Delta w_{i,t}|$ over a year. With a cost of κ per unit traded, the annual drag is about κu. So 200% turnover at 10 basis points costs 0.2 points a year, and at 50 basis points it costs a full point. Market impact is often modelled as growing with the square root of the order's share of daily volume (Tóth, Lempérière, Deremble, de Lataillade, Kockelkoren & Bouchaud, 2011). For individual-sized orders in liquid markets, impact is negligible and the spread dominates.

Why it appears here. §2.4 asks which anomalies survive costs. §7.1 and §7.7 treat costs as an implementation priority. §8.4 lists cost omission as a bias.

Deeper. Novy-Marx & Velikov (2016). Frazzini, Israel & Moskowitz (2018).

Part II — Markets, premia and portfolios. The economic vocabulary of §2 and §5.

A.8 Beta and the capital asset pricing model

The idea. Beta measures how much an asset moves with the market. The capital asset pricing model (CAPM) says that, in equilibrium, beta is the only risk that earns a reward, because all other risk can be diversified away. It is the benchmark theory against which "anomalies" are defined.

Formally. $\beta_i = \mathrm{Cov}(r_i, r_m)/\mathrm{Var}(r_m)$ for market excess return r_m. The CAPM predicts $\mathrm{E}[r_i] = \beta_i \, \mathrm{E}[r_m]$: expected excess return rises in a straight line with beta.

Why it appears here. The low-beta anomaly (§5.6) is the observation that the line is much flatter than predicted. Betting against beta (§5.6) levers low-beta stocks up to a beta of one. The regressions of §8.6 generalise the CAPM to several factors.

Deeper. Black (1972). Frazzini & Pedersen (2014).

A.9 Market efficiency, the random walk and the Grossman–Stiglitz paradox

The idea. A market is *informationally efficient* if prices already reflect the available information. Then no rule using that information earns more than a fair reward for the risk it takes. The crudest version, the *random walk*, says that future price changes cannot be predicted from past ones. Efficiency is a statement about returns *after* adjusting for risk. So any test of it is also a test of some model of what the fair reward is. Grossman and Stiglitz pointed out that perfect efficiency defeats itself. If prices already reflected everything, nobody would be paid to gather the information that makes them do so.

Formally. Fama (1970) distinguished three forms. Weak-form efficiency means that past prices are reflected. Semi-strong form covers all public information, and strong form all information. In the Grossman–Stiglitz equilibrium, the share of traders who pay to become informed adjusts until the expected gain from being informed just equals the cost of the information.

Why it appears here. §2.1 builds its account of which returns persist on this equilibrium. §3 describes how efficiency became the null hypothesis against which anomalies are tested.

Deeper. Fama (1970). Grossman & Stiglitz (1980). Pedersen (2015).

A.10 Risk premia, arbitrage, and the limits of arbitrage

The idea. A *risk premium* is the extra expected return paid for holding something that does badly in bad times, when losses are hardest to bear. It persists because the risk is real. An *arbitrage* is a riskless profit, and competition removes arbitrages quickly. Between the two lies the territory of §2: mispricings that are profitable on average but risky or costly to trade. The *limits of arbitrage* are the reasons that capital does not close them. The mispricing can widen before it closes. Arbitrageurs who manage other people's money are forced to retreat exactly when the opportunity is best.

Formally. In modern asset pricing, an asset's expected excess return is proportional to minus its covariance with a *stochastic discount factor* m, which is high in bad states: $\mathrm{E}[r] = -\mathrm{Cov}(m, r)/\mathrm{E}[m]$. Assets that pay off badly when m is high, in crises, must offer higher average returns.

Why it appears here. The four sources of §2.2 are distinguished by what the return is: a risk premium, a payment by a constrained party, or a mispricing protected by the limits of arbitrage.

Deeper. Cochrane (2011). Shleifer & Vishny (1997).

A.11 Benchmarks, tracking error and index flows

The idea. Most professional money is judged against a benchmark index. The risk that matters most to its managers is not losing money but lagging the benchmark. A position that might trail the index for years is dangerous to a manager's career, even when it is safe in absolute terms. So benchmarked money avoids it. Index funds add a mechanical flow. When a stock enters or leaves an index, every fund tracking that index must trade, on a date known in advance.

Formally. Tracking error is the standard deviation of the return difference from the benchmark, $\mathrm{sd}(r_p - r_b)$. An active portfolio's information ratio is its average return difference divided by its tracking error. That is the IR of §1.7.

Why it appears here. §2.2 and §2.5 attribute structural premia, such as the low-volatility anomaly and the index-inclusion effect, to benchmarking and index flows. §5.6 and §5.9 return to both.

Deeper. Baker, Bradley & Wurgler (2011). Greenwood & Sammon (2025).

A.12 Long–short portfolios and factors

The idea. A *factor* is the return of a portfolio built to isolate one characteristic. It is long the assets that have the characteristic, and short those that do not. The long and short sides roughly cancel the market's movement. What remains is the reward, if any, for the characteristic. In practice, the long–short position needs collateral, which earns the cash rate. So a factor's return is an excess return, and compounding it treats the position as fully collateralised.

Formally. The Fama–French value factor is $\mathrm{HML} = \frac{1}{2}(\text{small value} + \text{big value}) - \frac{1}{2}(\text{small growth} + \text{big growth})$. It comes from portfolios sorted on size and on book-to-market, which is the accounting book value of equity divided by its market value. Size (SMB) and momentum are built analogously. Momentum uses returns over months $t-12$ to $t-2$.

Value-weighted portfolios weight each stock by its market capitalisation, so the largest companies dominate. *Equal-weighted* portfolios give every stock the same weight, so the many small companies dominate. The smallest stocks, the *microcaps*, are numerous, illiquid and expensive to trade. So results that hold only with equal weighting often cannot be collected at scale. Replication studies therefore sort stocks using size breakpoints computed from NYSE-listed companies alone, which

keeps microcaps from defining the portfolios (§2.3). A long–short factor's long and short sides have similar exposure to the market. So the factor is roughly *market-neutral*: its return depends little on the market's direction.

Why it appears here. The worked numbers of §1.6, §2.3 and §2.6 use the Kenneth French library's market, size, value and momentum factors. §5 describes each family by its factor.

Deeper. Fama & French (1993). The documentation of the Kenneth French data library.

A.13 The information coefficient and the fundamental law

The idea. The information coefficient (IC) is the correlation between a forecast and what then happens. It is always small. What turns a small IC into a useful strategy is applying it to many *independent* bets. Their errors partly cancel, while their small edges add up.

Formally. For a jointly normal forecast and outcome with correlation ρ, the probability that they share a sign is $\frac{1}{2} + \arcsin(\rho)/\pi$. That is 51.6% at $\rho = 0.05$. Take BR independent bets a year, each contributing a small edge proportional to the IC and independent noise. The expected active return grows in proportion to BR, and its standard deviation in proportion to $\sqrt{\text{BR}}$. So their ratio grows as $\text{IC}\sqrt{\text{BR}}$. The transfer coefficient scales it down for constraints: $\text{IR} \approx \text{TC} \cdot \text{IC} \cdot \sqrt{\text{BR}}$.

Why it appears here. §1.7.

Deeper. Grinold (1989). Clarke, de Silva & Thorley (2002). Grinold & Kahn (2000).

A.14 Volatility targeting

The idea. Volatility clusters: a volatile month tends to be followed by another. So recent volatility is a usable forecast of near-term volatility, even though nothing forecasts returns well. Scaling each position inversely to its recent volatility keeps the risk it contributes roughly constant. It also cuts exposure when markets become turbulent.

Formally. A common volatility estimate $\hat{\sigma}_{i,t}$ is an exponentially weighted average of squared returns, $\hat{\sigma}_t^2 = \lambda\hat{\sigma}_{t-1}^2 + (1-\lambda)r_t^2$. The position is $w_{i,t} = \sigma^\star/\hat{\sigma}_{i,t}$ times the signal. (The `span=60` in the code of §7.6 corresponds to $\lambda = 1 - 2/61 \approx 0.967$.)

Why it appears here. The trend rule of §5.2, the sizing slot of §6.2, the code of §7.6 and the sizing discussion of §7.8.

Deeper. Moreira & Muir (2017). Harvey, Hoyle, Korgaonkar, Rattray, Sargaison & Van Hemert (2018). The chapter on trend-following.

A.15 The Kelly criterion

The idea. The Kelly criterion asks how much leverage maximises long-run compound growth. Too little leverage leaves growth on the table. Too much, and the volatility drag of A.1 overwhelms the extra return. The optimum balances the two.

Formally. Take a position with leverage L on a strategy with excess return μ and volatility σ. It has arithmetic mean $L\mu$ and variance $L^2\sigma^2$. So by A.1, its growth rate is about $g(L) = L\mu - \frac{1}{2}L^2\sigma^2$. Setting $g'(L) = 0$ gives $L^\star = \mu/\sigma^2$ and $g(L^\star) = \frac{1}{2}\mu^2/\sigma^2 = \frac{1}{2}\text{SR}^2$. Because g is a parabola, $g(L^\star/2) = \frac{3}{4}g(L^\star)$ and $g(2L^\star) = 0$.

Why it appears here. §7.8, where the last identity shows why full Kelly on an overestimated Sharpe ratio is dangerous.

Deeper. Kelly (1956). Thorp (2006). MacLean, Thorp & Ziemba (2011).

Part III — Statistics. The machinery of §8.

A.16 Standard errors, t-statistics, significance and power

The idea. Any estimate from finite data has sampling error. The standard error is its typical size. A t-statistic counts how many standard errors an estimate lies from zero. A *significance level* is the probability of declaring an effect when there is none. *Power* is the probability of detecting an effect that is there.

Formally. The mean of n independent observations with standard deviation σ has standard error $\sigma/\sqrt{n}$. t is the mean divided by the standard error. A two-sided test at 5% rejects when $|t| > 1.96$. If the true effect is δ standard errors, the test's power is about $\Phi(\delta - 1.96)$. So 80% power requires $\delta = 1.96 + 0.84 = 2.8$, since $\Phi(0.84) \approx 0.8$.

Why it appears here. The t-statistics of §1.6 and §2.3, and the table of years needed in §8.2.

Deeper. Section 4 of Foundations of Econometrics.

A.17 Regression, alpha and factor models

The idea. Regress a strategy's returns on the returns of portfolios that could be held directly. The regression asks how much of the strategy those portfolios reproduce. The *betas* are the exposures. The intercept, *alpha*, is the average return left over. Alpha is always alpha *relative to a chosen set of factors*. That is why the word means different things to different people (§6.4).

Formally. $\pi_t = \alpha + \sum_k \beta_k F_{k,t} + \varepsilon_t$, estimated by ordinary least squares. The residuals may be autocorrelated, as with overlapping or smoothed returns. Then the standard error of α must allow for it. Heteroskedasticity- and autocorrelation-consistent (HAC) standard errors widen it to reflect the correlation between nearby residuals. Newey–West standard errors are the usual choice.

Why it appears here. §8.6. The timing test of §1.6 is a one-variable regression of next month's return on the rule's position.

Deeper. Fama & French (1993). Carhart (1997). Section 3 of Foundations of Econometrics.

A.18 Multiple testing, and the best of M trials

The idea. Run enough tests, and some will pass by chance. Worse, the best of many noisy estimates is biased upward. Keeping only the winner means keeping the luckiest. Correcting for this means comparing the best result with the best result that luck alone would have produced, not with zero.

Formally. The expected maximum of M independent standard normal variables is approximately $(1-\gamma_E)\,\Phi^{-1}(1-1/M) + \gamma_E\,\Phi^{-1}(1-1/(Me))$, with $\gamma_E \approx 0.5772$ the Euler–Mascheroni constant. It grows roughly like $\sqrt{2\ln M}$. The formula comes from extreme value theory. Suitably rescaled, the maximum of many independent normal variables approaches a Gumbel distribution, whose mean involves the Euler–Mascheroni constant. The crude Bonferroni correction tests each of M hypotheses at level α/M. For $M = 100$ at 5% two-sided, that means $|t| > 3.5$.

Why it appears here. §8.3, and the opening example of §1.1.

Deeper. Bailey & López de Prado (2014). Harvey, Liu & Zhu (2016).

A.19 The probabilistic and deflated Sharpe ratios

The idea. These are two corrections to the naive question "is the Sharpe ratio above zero?". The *probabilistic* Sharpe ratio asks how likely it is that the true Sharpe ratio exceeds a benchmark, given the length of the sample and the shape of the returns. So it penalises short histories, negative skew and fat tails. The *deflated* Sharpe ratio sets that benchmark at the best Sharpe ratio that luck alone would have produced across all the variants tried.

Formally. Take a per-period estimate $\widehat{\mathrm{SR}}$ from n observations with skewness γ_3 and kurtosis γ_4. The probabilistic Sharpe ratio against a benchmark SR^* is

$$\Phi\left(\frac{(\widehat{\mathrm{SR}} - \mathrm{SR}^*)\sqrt{n-1}}{\sqrt{1 - \gamma_3\widehat{\mathrm{SR}} + \frac{\gamma_4 - 1}{4}\widehat{\mathrm{SR}}^2}}\right).$$

The deflated Sharpe ratio evaluates it with SR^* set to the expected maximum of the previous entry. That maximum is scaled by the standard deviation of the Sharpe ratios across the trials.

Why it appears here. The denominator is the non-normal standard error of §8.2. §8.3 uses the deflated version.

Deeper. Bailey & López de Prado (2012). Bailey & López de Prado (2014).

A.20 The bootstrap, and the reality check

The idea. Sometimes the sampling distribution of a statistic is hard to derive. The bootstrap simulates it by resampling the available data. Resampling blocks of consecutive observations, not single ones, preserves short-range dependence.

White's *reality check* bootstraps the *best* performance across every strategy tried, not one strategy's performance. That is the right null for a search.

Formally. Draw B resamples of the return history and recompute the statistic on each. Use the empirical distribution of the B values as its sampling distribution. For the reality check, the statistic is the maximum, across all M rules, of each rule's average excess performance.

Why it appears here. §8.3.

Deeper. White (2000). Hansen (2005). Sullivan, Timmermann & White (1999).

A.21 Holdouts, walk-forward testing and purged cross-validation

The idea. A model must be judged on data it was not fitted to. A *holdout* sets aside the most recent data. *Walk-forward* testing repeatedly fits on the past and tests on the next period. *Cross-validation* rotates which part of the data is held out. In finance it must be *purged*: observations whose outcome windows overlap the test set are removed from training. Otherwise information leaks from test to train. The probability of backtest overfitting of §8.3 is estimated by a symmetric version of this rotation.

Formally. In k-fold cross-validation, the sample is split into k blocks. Each block serves once as the test set for a model fitted on the others. Purging drops training observations whose labels overlap the test block's time span. An *embargo* also drops a short period after it.

Why it appears here. §8.3 and §8.7.

Deeper. López de Prado (2018). The chapter on market regimes.

A.22 The CUSUM

The idea. The CUSUM is a running tally of how far a series has fallen short of what was expected. It resets to zero whenever the series is ahead. When the tally exceeds a threshold, the detector concludes that the mean has shifted. The threshold trades the rate of false alarms against the delay in catching a real change.

Formally. To detect a fall in the mean of x_t from μ_0 to μ_1, set $S_0 = 0$ and $S_t = \max\left(0,\ S_{t-1} + \frac{1}{2}(\mu_0 + \mu_1) - x_t\right)$. Raise an alarm when $S_t > h$. §8.8 uses $\mu_1 = 0$, meaning the strategy has died, and measures h in monthly standard deviations.

Why it appears here. §8.8.

Deeper. Page (1954).

Part IV — Instruments and markets. The vehicles of §5 and §7.

A.23 Futures: multipliers, margin, rolls and carry

The idea. A futures contract is an agreement to buy or sell something at a fixed price on a future date. Gains and losses are settled daily. So the margin posted is a performance bond, not a down payment, which is why futures embed leverage. Contracts expire, so holding a position means *rolling* it: closing the expiring contract and opening the next. Commodity trading advisers (CTAs) are the regulated managers who run futures strategies for clients. Trend-following is their dominant style.

Formally. A contract's notional value is its multiplier times the price. For a micro S&P 500 contract, that is 5 dollars times the index. The *basis* is the gap between the futures price F and the spot price S. *Carry* is $(S - F)/F$, the return from the futures price converging to an unchanged spot price. When futures trade above spot, the curve is in *contango*. When they trade below, it is in *backwardation*. Crypto exchanges also list *perpetual* futures, which never expire. Periodic *funding* payments between longs and shorts keep their price near spot. Those payments are the crypto carry of §5.5.

Why it appears here. Trend and carry (§5.2, §5.5). Vehicles and minimum account size (§7.3, §7.4). Continuous-series construction (§7.5).

Deeper. Koijen, Moskowitz, Pedersen & Vrugt (2018). The chapter on trend-following.

A.24 Options, put–call parity and implied volatility

The idea. A call option is the right to buy at a fixed *strike* price. A put is the right to sell. Their prices depend on how volatile the underlying is expected to be. So an option price can be quoted as a volatility: the *implied volatility* that makes a pricing model reproduce the market price. Combining options and the underlying gives payoffs of almost any shape. Some combinations have identical payoffs, and so must have identical prices.

Formally. With strike X, at expiry a call pays $\max(S - X, 0)$ and a put pays $\max(X - S, 0)$. A portfolio of a call plus cash worth X at expiry pays $\max(S, X)$. So does a put plus the stock. Equal payoffs imply equal prices: $C + Xe^{-r_f \tau_e} = P + S$, which is put–call parity. Several related terms appear in the main text:

- An option's *delta* is the sensitivity of its price to the underlying's price.
- *Delta hedging* holds an offsetting position in the underlying, adjusted as delta changes. What remains is a bet on volatility, not on direction.
- A *straddle* is a call plus a put at the same strike. It pays on large moves either way. A *lookback* straddle pays the full range between the highest and lowest prices.
- A *covered call* is a stock position with a call sold against it.
- A *cash-secured put* is a short put backed by cash equal to the strike.
- *Same-day* options, often called 0DTE, expire on the day they are traded.

Why it appears here. Volatility selling (§5.7) and the equivalences of §6.3.

Deeper. The chapters on implied volatility and dealer hedging.

A.25 Convex and concave strategies

The idea. A strategy is *convex* in the market if it gains more from large moves than it loses from small ones. That is the shape of an option buyer's payoff. A strategy is *concave* if the reverse holds, which is the shape of an option seller's payoff. Many strategies that hold no options still have one of these shapes, because of how they trade. Buying as prices rise and selling as they fall produces convexity. Rebalancing back to fixed weights sells after rises and buys after falls, which produces concavity.

Formally. Perold & Sharpe (1988) compare three rules for a stock–cash portfolio. Buy-and-hold is linear in the stock price. Constant-mix rebalancing is concave: it gains when prices oscillate, and loses when they trend. Portfolio insurance raises the stock weight after rises, and is convex: it gains in trends and loses in oscillations. A convex strategy is in effect long volatility, and a concave one is short it.

Why it appears here. Rebalancing in §5.1. In §5.2 and §6.3, trend-following as a long straddle, and merger arbitrage and carry as short puts.

Deeper. Perold & Sharpe (1988). Fung & Hsieh (2001).

A.26 The variance risk premium and the VIX

The idea. Option prices imply a forecast of future volatility. On average, that forecast has been higher than the volatility that followed. The difference is what buyers pay for protection and what sellers earn. The VIX index summarises the S&P 500's implied volatility over the next 30 days. VIX futures let investors trade it directly.

Formally. Option prices embed *risk-neutral* probabilities. These weight bad outcomes more heavily than their real-world frequency, because investors pay more for payoffs in bad states. The variance risk premium is the difference between two quantities. One is the risk-neutral expectation of future realised variance, which can be computed from a strip of option prices, as the VIX does. The other is the realised variance itself. VIX futures usually trade above the spot index (contango). So short positions earn a roll return in calm markets, and lose heavily when volatility spikes.

Why it appears here. §5.7.

Deeper. Carr & Wu (2009). Augustin, Cheng & Van den Bergen (2021). The chapter on implied volatility.

A.27 ETFs and leveraged ETFs

The idea. An exchange-traded fund holds a basket of assets and trades like a single stock. A creation and redemption mechanism keeps its price close to the basket's value. A *leveraged* ETF resets to a fixed multiple of its index every day. That makes its long-run return depend on the path the index takes, not just on where the index ends.

Formally. With daily rebalancing, the fund's instantaneous return is L times the index's. By A.1, its log return grows at $L\mu - \frac{1}{2}L^2\sigma^2$. L times the index's log return grows at $L\mu - \frac{1}{2}L\sigma^2$. The difference, $-\frac{1}{2}(L^2 - L)\sigma^2$ per year, is the drag of §7.3.

Why it appears here. §7.3.

Deeper. Avellaneda & Zhang (2010).

A.28 Orders, auctions and payment for order flow

The idea. A *market order* trades immediately at the best available price. A *limit order* trades only at a stated price or better, and may not trade at all. A *marketable limit order* is priced to trade at once, but caps the price. Exchanges set the official closing price in a *closing auction*, which market-on-close and limit-on-close orders join. Most US retail stock orders are routed to wholesale market makers. These execute the orders off-exchange and pay the broker for the flow.

Formally. The *bid–ask spread* is the gap between the best price to sell and the best price to buy. A round trip at the quotes costs one spread. *Price improvement* is execution inside the quotes. *Implementation shortfall* is the difference between the price a strategy assumed and the price it got.

Why it appears here. §7.7.

Deeper. Schwarz, Barber, Huang, Jorion & Odean (2025). Bryzgalova, Pavlova & Sikorskaya (2023).

Part V — US rules. The constraints of §7, as of September 2026.

A.29 US taxation of trading gains

The idea. US tax treatment depends on how long a position was held, and on what kind of instrument it is. Short holding periods are taxed like wages. Long ones are taxed at lower rates. Some derivatives are taxed at a fixed blend, regardless of holding period. Deferring tax is itself valuable, because untaxed gains keep compounding.

Formally. Gains on positions held a year or less are taxed at ordinary rates. Longer ones are taxed at long-term rates. Section 1256 contracts are taxed 60% long-term and 40% short-term, marked to market each year-end. The wash-sale rule disallows a loss if a substantially identical security is bought within 30 days before or after the sale. Take a pre-tax return g over Y years and a tax rate τ. Realising every year compounds to $\big(1 + g(1-\tau)\big)^Y$. Deferring to the end gives $1 + \big((1+g)^Y - 1\big)(1-\tau)$. These are the two formulas behind the table in §7.2.

Why it appears here. §7.2.

Deeper. IRS Publication 550 and Topic 409. 26 U.S.C. section 1256 (§4.12).

A.30 Margin, Regulation T, and day-trading rules

The idea. Buying on margin means borrowing from the broker against the securities bought. Regulators set minimum collateral levels. Brokers can set higher ones, and can sell positions when an account falls below them.

Formally. Regulation T sets initial margin for stocks at 50% of the purchase price. FINRA's rules set maintenance margin at no less than 25%. Some brokers offer *portfolio margin* to larger accounts. It sets requirements from a stress test of the whole account instead, and usually allows more leverage on hedged positions. Until June 2026, FINRA required accounts of "pattern day traders" to hold at least 25,000 dollars. A pattern day trader was, broadly, one making four or more day trades within five business days. The replacement standard ties required equity to intraday exposure instead.

Why it appears here. §7.4.

Deeper. FINRA (2026).

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